Bounded Self-Referential Dynamics in a 4D Soft-Wall ODE System

Appendix B v1.0: Post-Hopf Relaxation Cycles, Ψ-Memory Persistence, Saddle-Node Cycle Bifurcation, Variational Floquet Analysis, and Pointwise Boundary Algebra

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Abstract

We study a dimensionless, unit-normalized 4D ordinary differential equation system $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ that extends the 3D hyperbolic-tangent soft-wall model of Appendix A v1.5 (DOI: 10.5281/zenodo.23534449) by coupling a slow-decaying activity coordinate $\Psi \in [0,1]$ driven by the smooth (C^∞) quadratic feedback ratio $R_{\text{self}} = \text{fb}^2/(\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ and an effective input flux $\mathcal{P}_{\text{eff}} = \mathcal{P}_0 + \Lambda_{\text{eff}}\gamma + \lambda_{\text{lucid}}\Psi$ ($\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}}$). In Lean 4 (`ExistenceVerification.lean`, 17 theorems, zero `sorry`), we machine-check the pointwise algebraic sign conditions on all eight faces of the compact box $\mathcal{B}_4 = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max,4D}] \times [0,1]$ together with the pointwise linear comparison bounds $d\gamma/dt \geq -b\gamma$ and $d\dot{I}/dt \geq f\mathcal{P}_0 - (g + eM_H/\epsilon_I)\dot{I}$, from which continuous-time forward invariance of $\mathcal{B}_4$, finite-time positivity $\gamma(t) \geq \gamma(0)e^{-bt} > 0$, and uniform asymptotic floors $\liminf_{t\to\infty} \dot{I}(t) \geq \dot{I}_{\text{floor}}(\mathcal{P}_0) > 0$ and $\liminf_{t\to\infty} \gamma(t) \geq \gamma_{\text{floor}}(\mathcal{P}_0) > 0$ (for $\mathcal{P}_0 > 0$) follow by classical Nagumo and Grönwall comparison arguments.

Because R_{self} is quadratic in $\text{fb} = e\tanh(\dot{I}/\epsilon_I)\dot{H}_{\text{obs}}$, both R_{self} and ∇R_{self} vanish at any steady state ($\dot{H}^*_{\text{obs}} = 0$), so λ_{lucid} decouples by construction from the equilibrium Jacobian J_4. For $\mathcal{P}_0 > 0$, we prove global monotonicity $de_c/d\mathcal{P}_0 > 0$ of the $\epsilon \to 0^+$ open-loop Hopf threshold on $\mathcal{P}_0 > 10/99$, derive the closed-loop threshold $e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}})$, and evaluate transverse Floquet multipliers on the Poincaré section $\Sigma = \{\gamma = 2.0, \dot{\gamma} > 0\}$ by integrating the 20D first-variation monodromy equations. At zero external drive ($\mathcal{P}_0 = 0$) with $0 \leq \Lambda_{\text{eff}} < 1$, no positive interior equilibrium or Hopf bifurcation exists and the origin $(0,0,0,0)$ is locally asymptotically stable; a 2×2 ablation and continuation in λ_{lucid} reveal that a memoryless proportional loop ($\Lambda_{\text{eff}} = 0.80, \lambda_{\text{lucid}} = 0$) collapses to the origin because $\gamma(t)$ drops to $\sim 3.6 \times 10^{-4}$ at the cycle trough, whereas slow Ψ-memory ($\Psi_{\min} \approx 0.731, \lambda_{\text{lucid}}\Psi \geq 0.219$) bridges the trough and sustains a bistable periodic orbit born via a saddle-node bifurcation of limit cycles at $\lambda_{\text{lucid,SNLC}}(0) = 0.080341$ (lowered by 27.8% to $\lambda_{\text{lucid,SNLC}}(0.80) = 0.057974$ with a 9.14× boost in period-averaged flux when coupled with $\Lambda_{\text{eff}} = 0.80$).

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1 What Does It Mean for a Finite System to Exist?

When we ask what it means for something to *exist*, we are often pulled toward two extremes. At one extreme, we think of a stone sitting quietly on the ground—static, unchanging, and indifferent to time. At the other extreme, people reach for grand, sweeping mysteries: a physical "Theory of Everything" uniting quantum gravity and particle fields, or big philosophical claims about consciousness and the soul.

In my work across the `Reggae-shark-universe-` research program, I have learned to distrust both extremes. A set of four differential equations is not the Standard Model of particle physics, and a number crossing 0.50 on a computer screen does not give a machine inner feelings or human awareness. Pretending that a mathematical model does more than what is actually proved in its equations violates the rule that guides everything I build: **Build, Attack, Falsify, Preserve**.

Yet between a silent stone and the whole cosmos lies a simpler, deeply human question that mathematics *can* answer with complete honesty: *How does a finite, self-reflecting system hold itself together over time without burning out or falling apart?*

When I first watched the unregularized equations of Physics v1.0 crash at $t \approx 55$ (DOI: 10.17605/OSF.IO/N3DQU), it looked at first like a purely technical flaw in a polynomial differential equation. Once the system's internal feedback gain e was turned up past a critical tipping point e_c, the positive first Lyapunov coefficient ($l_1 = \frac{305328816}{1498181485}\sqrt{13/2} \approx +0.519589 > 0$) pushed the system away from its steady balance without giving it a stable nearby orbit to catch it. Driven by an unchecked memory-erasure term $-\dot{I}_{\text{erase}}$ and an unchecked self-feedback loop $e\,\dot{H}_{\text{obs}}$, the system's internal uncertainty $H_{\text{obs}}(t)$, processing speed $\dot{I}(t)$, and reconstruction capacity $\gamma(t)$ plunged straight through zero into negative numbers.

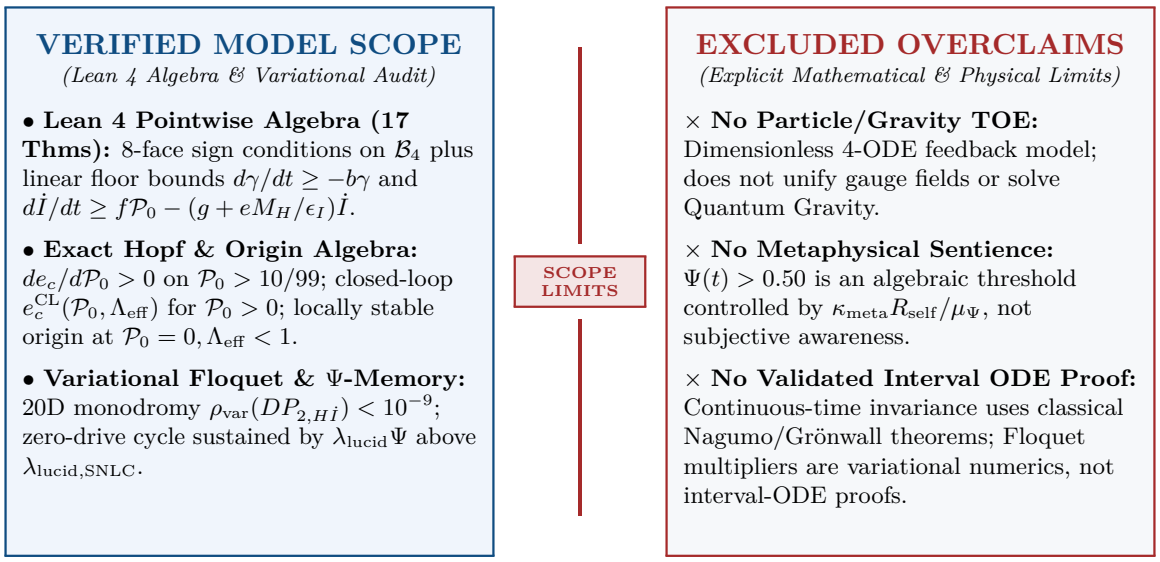


Figure 1: Scope boundary of Appendix B v1.0, separating the machine-checked pointwise boundary algebra, symbolic bifurcation analysis, and 20D variational monodromy numerics from unverified physical, philosophical, or interval-arithmetic overclaims.

As I worked through Appendix A v1.1 to v1.5 (DOI: 10.5281/zenodo.23534449), I realized that this crash was not just a quirk of algebra. It was pointing to a basic law of survival for any bounded system. In cybernetics and viability theory—from W. Ross Ashby's *Design for a Brain* (1952) and Humberto Maturana and Francisco Varela's idea of *autopoiesis* (self-maintenance, 1972) to Jean-Pierre Aubin's *Viability Theory* (1991)—to **exist** as a finite observer means staying inside a livable, bounded region of state space for all future time $t \geq 0$:

1. If a system's internal variables blow up to infinity ($H_{\text{obs}} \to \infty$ or $\dot{I} \to \infty$), its finite hardware overloads and breaks.

2. If its capacity to rebuild its internal model drops to zero or goes negative ($\gamma \leq 0$), the system loses its ability to track the world at all—it ceases to function.

Seen this way, the two smooth hyperbolic-tangent "soft walls" introduced in Appendix A v1.5—$\Psi_{\epsilon_H}(H_{\text{obs}}) = \tanh(H_{\text{obs}}/\epsilon_H)$ on the memory-erasure channel and $\Phi_{\epsilon_I}(\dot{I}) = \tanh(\dot{I}/\epsilon_I)$ on the self-feedback channel—are not just convenient mathematical patches. They represent the **natural physical limits** that allow a self-reflecting system to survive when its internal sensitivity e rises past its steady tipping point e_c.

1.1 Why Self-Reflection Without Natural Limits Tears a System Apart

Think for a moment about how a purely passive filter works when self-reflection is turned off ($e = 0$). A passive filter simply takes in outside information $\mathcal{P}_{\text{input}}$, processes it at a steady rate $\dot{I}^* = f\mathcal{P}_{\text{input}}/g$, maintains a steady capacity $\gamma^* = K\gamma_{\max}/(b\gamma_{\max} + K)$, and settles at a comfortable level of internal entropy $H^*_{\text{obs}} = H_{\max} - \dot{I}_{\text{erase}}/(c\gamma^*)$. It is completely stable, but it is also blind to its own inner changes: it never reacts to how fast its own uncertainty is rising or falling.

The moment we give the system *self-reflection*—feeding the rate of change of its own uncertainty dH_{obs}/dt back into its processing acceleration $d\dot{I}/dt$ with gain $e > 0$—everything changes. At modest levels ($0 < e < e_c$), this inner sensitivity makes the system responsive while keeping its balance. But if the gain e is too high, or if the outside world goes quiet ($\mathcal{P}_0$ drops) while internal sensitivity e stays high, the system starts over-reacting to its own fluctuations. Without natural limits, a sudden drop in entropy ($\dot{H}_{\text{obs}} < 0$) pulls processing speed $\dot{I}(t)$ below zero, which then drags capacity $\gamma(t)$ below zero. In plain words: the system tries to clear out memory it no longer has, and tries to draw power from an engine that has already stopped.

To survive strong self-reflection or quiet periods when outside input drops, any real system needs two simple, common-sense safeguards:

- **You Cannot Erase an Empty Page ($\Psi_{\epsilon_H}(H_{\text{obs}}) = \tanh(H_{\text{obs}}/\epsilon_H)$):** As internal entropy H_{obs} approaches zero, memory erasure must smoothly slow down and stop at zero.
- **You Cannot Amplify a Stopped Engine ($\Phi_{\epsilon_I}(\dot{I}) = \tanh(\dot{I}/\epsilon_I)$):** As processing rate $\dot{I}$ approaches zero, internal derivative feedback must smoothly let go until incoming drive builds processing back up.

Once these two soft walls are in place, crossing the stability threshold e_c no longer kills the system. Instead of crashing through zero, the system is gently caught by its lower boundaries and enters a stable, repeating rhythm—a bounded cycle of resting and rebuilding inside a closed four-dimensional box.

1.2 Three Questions This Paper Answers

Once we know the system cannot escape its box, three clear questions open up before us:

- **Question 1 (What happens when the outside world goes quiet?):** If an observer has a fixed internal sensitivity ($e = 3.60$) that is stable during strong outside input ($\mathcal{P}_0 = 5.0$), what happens to its balance when outside input drops ($\mathcal{P}_0 = 4.0$ or $\mathcal{P}_0 = 1.0$) while internal memory clearing $\dot{I}_{\text{erase}}$ keeps running?
- **Question 2 (What is the difference between noticing an inner rhythm and steering it?):** Suppose we add a fourth variable $\Psi(t) \in [0, 1]$ that tracks how much of the system's activity comes from its own internal feedback loop versus the outside world. How does $\Psi(t)$ behave at a calm steady state versus during an oscillation? And what changes when $\Psi(t)$ is merely an observer readout ($\lambda_{\text{lucid}} = 0$) versus when it feeds back to steer the processing channel ($\lambda_{\text{lucid}} > 0$)?

- **Question 3 (What actually keeps the rhythm alive when external drive is shut off completely at $\mathcal{P}_0 = 0$?):** When $\mathcal{P}_0 = 0$ and $0 \leq \Lambda_{\text{eff}} < 1$, no positive interior equilibrium or Hopf bifurcation exists, and the silent origin $(0, 0, 0, 0)$ is locally asymptotically stable. If we rely only on instantaneous proportional feedback $\Lambda_{\text{eff}}\gamma(t)$ ($\lambda_{\text{sim}} = 7.20, \lambda_{\text{lucid}} = 0$), the oscillation *dies to the origin*, because $\gamma(t)$ drops to $\sim 3.6 \times 10^{-4}$ during the deep erasure trough and starves $\dot{I}(t)$ of input drive. By contrast, because $\Psi(t)$ decays slowly ($\mu_\Psi = 0.60$), it stays above $\Psi_{\min} \approx 0.731$ across the trough so that $\lambda_{\text{lucid}}\Psi(t) \geq 0.219$ acts as a **slow capacitive memory reservoir** that re-ignites the next burst. Continuation in λ_{lucid} shows that this zero-drive cycle is born via a **saddle-node bifurcation of limit cycles (SNLC)** at $\lambda_{\text{lucid,SNLC}}(0) = 0.080341$, which is lowered by 27.8% to $\lambda_{\text{lucid,SNLC}}(0.80) = 0.057974$ (with a 9.14× boost in period-averaged flux $\overline{\mathcal{P}}_{\text{eff},T}$) when $\Lambda_{\text{eff}} = 0.80$ amplifies the recovery stroke.

A Note on Plain Language, Unit Normalization, and Mathematical Scope: Equations (4)–(7) define an **illustrative, dimensionless 4D dynamical system** with unit-normalized structural constants ($a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1$), constructed to explore the mathematical mechanics of bounded self-referential feedback rather than to serve as an empirical model of physical or biological systems. Phrases like *waking-type steady state*, *offline relaxation cycle*, *high-Ψ readout*, and *autopoietic closure* are used solely as descriptive labels for specific bifurcation regimes of these differential equations. In particular, because $\Psi^* = 0$ at every steady state ($\dot{H}^*_{\text{obs}} = 0$), $\Psi(t)$ measures departure from equilibrium driven by internal feedback fb^2 relative to $(f\mathcal{P}_0)^2 + \sigma_0^2$, and crosses 0.50 simply when the feedback ratio R_{self} is large enough that $\kappa_{\text{meta}}R_{\text{self}}(1 - \Psi)$ outpaces decay $\mu_\Psi\Psi$.

2 The Smooth (C^∞) 4D Dynamical System

If we want our mathematical proofs, our symbolic Jacobian matrices, and our computer simulations to tell the exact same story without hidden glitches, every equation in the model must be smooth (C^∞) everywhere. If one uses a sharp absolute-value ratio $|\text{fb}|/(|\text{fb}| + f\mathcal{P}_0)$ to measure internal feedback, that formula has a sharp corner (a V-shaped kink) right at $\text{fb} = 0$ ($\dot{H}_{\text{obs}} = 0$). Because every steady state sits at $\dot{H}_{\text{obs}} = 0$, and every oscillation crosses $\dot{H}_{\text{obs}} = 0$ twice per cycle, a sharp corner breaks the derivative at equilibrium and causes different numerical solvers to drift apart.

By using a **single, smooth quadratic ratio** $R_{\text{self}} = \text{fb}^2/(\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ across every part of this work (Lean 4 proofs, `SymPy` algebra, `SciPy` solvers, and the interactive simulator), the kink disappears completely: the curve is flat and smooth right across $\text{fb} = 0$.

Definition 1 (The Smooth 4D Bounded Existence System)**.** Let the state of the system at time $t \geq 0$ be $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t)) \in \mathbb{R}^4$, where:

- $\gamma(t) \in [0, \gamma_{\max}]$ is the dimensionless internal reconstruction capacity,
- $H_{\text{obs}}(t) \in [0, H_{\max}]$ is the dimensionless internal entropy or uncertainty coordinate,
- $\dot{I}(t) \in [0, \dot{I}_{\max,4D}]$ is the dimensionless processing rate coordinate,
- $\Psi(t) \in [0, 1]$ is the dimensionless slow-memory / self-feedback coordinate (motivated by the self-modeling discriminator of Program 02, DOI: 10.17605/OSF.IO/8CFQW), which tracks non-equilibrium derivative-feedback activity and acts as a slow capacitive buffer when $\lambda_{\text{lucid}} > 0$.

For strictly positive structural constants $a, b, c, f, g, \mathcal{E}_{\text{avail}}, \gamma_{\max}, H_{\max}, \dot{I}_{\text{erase}}, \epsilon_H, \epsilon_I, \sigma_0, \mu_\Psi > 0$ and non-negative control settings $e, \mathcal{P}_0, \lambda_{\text{sim}}, r_{\text{LRSC}}, \lambda_{\text{lucid}}, \kappa_{\text{meta}} \geq 0$, define the effective input drive $\mathcal{P}_{\text{eff}}(\gamma, \Psi)$, the internal derivative-feedback signal $\text{fb}(\gamma, H_{\text{obs}}, \dot{I})$, and the smooth quadratic self-feedback ratio

$R_{\text{self}}(\gamma, H_{\text{obs}}, \dot{I})$ by:

$$\mathcal{P}_{\text{eff}}(\gamma, \Psi) = \mathcal{P}_0 + \underbrace{\lambda_{\text{sim}}\, r_{\text{LRSC}}}_{\Lambda_{\text{eff}}}\, \gamma + \lambda_{\text{lucid}}\, \Psi, \tag{1}$$

$$\text{fb}(\gamma, H_{\text{obs}}, \dot{I}) = e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\left[c\gamma(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}} \tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right)\right], \tag{2}$$

$$R_{\text{self}}(\gamma, H_{\text{obs}}, \dot{I}) = \frac{\text{fb}(\gamma, H_{\text{obs}}, \dot{I})^2}{\text{fb}(\gamma, H_{\text{obs}}, \dot{I})^2 + (f\mathcal{P}_0)^2 + \sigma_0^2}. \tag{3}$$

The time evolution of $x(t)$ is governed by the smooth (C^∞) autonomous system $dx/dt = F_{\text{exist}}(x)$:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{4}$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}} \tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{5}$$

$$\frac{d\dot{I}}{dt} = e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma, \Psi) - g\dot{I}, \tag{6}$$

$$\frac{d\Psi}{dt} = \kappa_{\text{meta}}\, R_{\text{self}}(\gamma, H_{\text{obs}}, \dot{I})\,(1 - \Psi) - \mu_\Psi \Psi. \tag{7}$$

Remark 2 (Two Simple Facts About the Parameters Λ_{eff} and σ_0)**.** Looking at Eqs. (1)–(3), two practical points are worth noting right away:

1. **Only the Product $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}}$ Acts on the Equations:** In Eq. (1), λ_{sim} and r_{LRSC} appear only multiplied together as $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}}$. Any two pairs $(\lambda_{\text{sim}}, r_{\text{LRSC}})$ with the same product produce the exact same motion. Setting $r_{\text{LRSC}} = K_{0.001}/M = 11/99 = 1/9$ from Program 04 (DOI: 10.17605/OSF.IO/NM5BW) simply fixes the scale so that $\lambda_{\text{sim}} = 7.20$ corresponds to an effective loop gain of $\Lambda_{\text{eff}} = 0.80$.

2. **Why $\sigma_0 > 0$ Is in the Denominator:** Including the small constant $\sigma_0^2 > 0$ ($\sigma_0 = 0.05$) in the denominator of R_{self} guarantees that $\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2 \geq \sigma_0^2 > 0$ everywhere. Even when outside input is completely turned off ($\mathcal{P}_0 = 0$) and $\text{fb} = 0$, we never divide by zero, and R_{self} stays smooth everywhere.

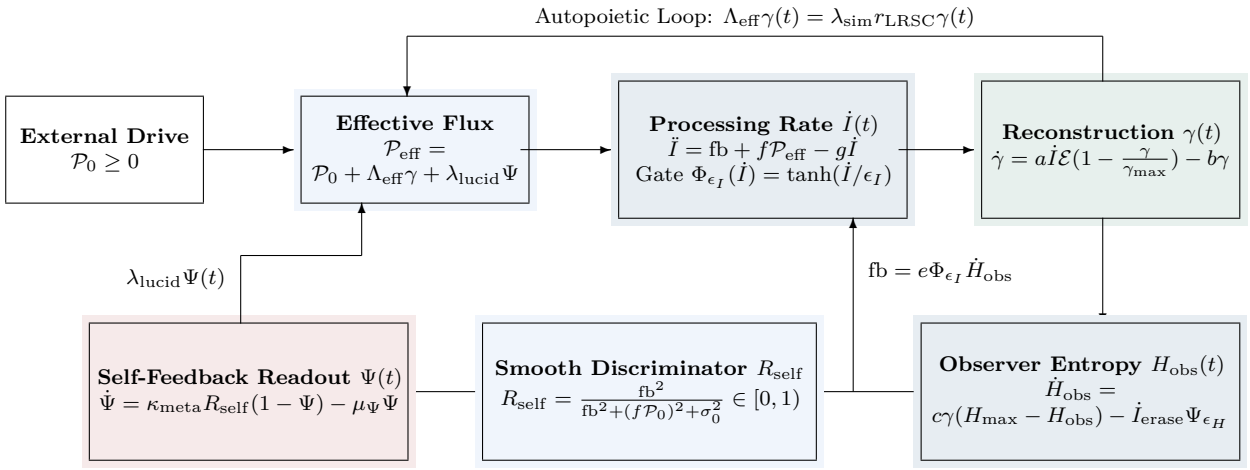


Figure 2: Block architecture of the smooth (C^∞) 4D system Eqs. (4)–(7). At any steady state ($\dot{H}^*_{\text{obs}} = 0$), $\text{fb} = 0$ and $\nabla R_{\text{self}} = 0$, decoupling λ_{lucid} from linear stability while $\Lambda_{\text{eff}}\gamma$ shifts the closed-loop equilibrium and Hopf threshold.

2.1 Why the Smooth Quadratic Ratio $R_{\text{self,sq}}$ Matters

To see clearly why squaring the feedback signal in $R_{\text{self,sq}} = \text{fb}^2/(\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ makes such a big difference compared to an absolute-value ratio $R_{\text{self,abs}} = |\text{fb}|/(|\text{fb}| + f\mathcal{P}_0)$, let us look at their slopes right near $\text{fb} = 0$:

$$\frac{\partial R_{\text{self,abs}}}{\partial \text{fb}} = \frac{\text{sgn}(\text{fb})\, f\mathcal{P}_0}{(|\text{fb}| + f\mathcal{P}_0)^2} \xrightarrow{\text{fb}\to 0^{\pm}} \pm\frac{1}{f\mathcal{P}_0} \neq 0, \qquad \frac{\partial R_{\text{self,sq}}}{\partial \text{fb}} = \frac{2\,\text{fb}\,\left[(f\mathcal{P}_0)^2 + \sigma_0^2\right]}{\left[\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2\right]^2} \xrightarrow{\text{fb}\to 0} 0. \quad (8)$$

With an absolute value, the slope jumps suddenly from $-1/(f\mathcal{P}_0)$ to $+1/(f\mathcal{P}_0)$ at $\text{fb} = 0$, so the system does not even have a well-defined Jacobian matrix at its own steady state. With the quadratic ratio $R_{\text{self,sq}}$, the slope approaches zero smoothly from both sides ($\nabla R_{\text{self}}|_* = 0$), giving a clean block-triangular Jacobian matrix at equilibrium and allowing our independent numerical integrators (`LSODA` and `DOP853`) to agree to roughly ten decimal places ($\sim 10^{-9}$ at $\text{rtol} = 10^{-10}$ and $\sim 10^{-11}$ at $\text{rtol} = 10^{-12}$).

3 Staying Inside the Box & Strict Interior Positivity

There is an important difference between saying a variable *never goes negative* ($\gamma(t) \geq 0$ on the closed box $\mathcal{B}_4$) and proving that it *stays strictly positive* ($\gamma(t) > 0, H_{\text{obs}}(t) > 0, \dot{I}(t) > 0$ for all finite time $t \geq 0$ whenever it starts inside the box). Because the deep troughs of our relaxation cycles dip quite close to the floor ($\gamma_{\min} \approx 0.0017$ in Regime IVa and $\gamma_{\min} \approx 0.0004$ in the periodic basin at $\mathcal{P}_0 = 0$), we need both: first, an 8-face boundary proof showing that every wall of the 4D box $\mathcal{B}_4$ points inward, and second, an exponential Grönwall lower bound proving that $\gamma(t)$, $H_{\text{obs}}(t)$, and $\Psi(t)$ cannot touch zero in any finite time.

Theorem 3 (8-Face Global Forward-Invariance of $\mathcal{B}_4$ and Grönwall Interior Positivity — `ANALYTIC_THEOREM`)**.** *Let* $a, c, e, \mathcal{E}_{\text{avail}}, \mathcal{P}_0, \lambda_{\text{sim}}, r_{\text{LRSC}}, \lambda_{\text{lucid}}, \kappa_{\text{meta}} \geq 0$ *and* $b, f, g, \gamma_{\max}, H_{\max}, \dot{I}_{\text{erase}}, \epsilon_H, \epsilon_I, \sigma_0, \mu_\Psi > 0$. *Define the maximum entropy rate* $M_H = c\gamma_{\max}H_{\max} + \dot{I}_{\text{erase}}$, *the maximum effective input drive* $\mathcal{P}_{\text{eff,max}} = \mathcal{P}_0 + \lambda_{\text{sim}} r_{\text{LRSC}} \gamma_{\max} + \lambda_{\text{lucid}}$, *and the 4D processing ceiling:*

$$\dot{I}_{\max,4\text{D}} = \frac{eM_H + f\mathcal{P}_{\text{eff,max}}}{g}. \quad (9)$$

Then:

1. ***8-Face Compact Box Invariance (Including*** $\mathcal{P}_0 = 0$***):*** *The compact hyper-box* $\mathcal{B}_4 = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max,4\text{D}}] \times [0, 1]$ *satisfies Nagumo's inward-pointing boundary conditions on all eight lower and upper faces and is globally forward-invariant for all* $t \geq 0$. *When* $\mathcal{P}_0 > 0$, *the lower processing face* $\dot{I} = 0$ *pushes strictly inward* ($d\dot{I}/dt\big|_{\dot{I}=0} \geq f\mathcal{P}_0 > 0$).

2. ***Strict Interior Positivity via Grönwall Comparison Bounds:*** *For any trajectory starting in the interior* $x(0) \in \text{int}(\mathcal{B}_4) = (0, \gamma_{\max}) \times (0, H_{\max}) \times (0, \dot{I}_{\max,4\text{D}}) \times (0, 1)$, *the coordinates obey the differential inequalities:*

$$\frac{d\gamma}{dt} \geq -b\gamma(t), \qquad \frac{dH_{\text{obs}}}{dt} \geq -\left(c\gamma_{\max} + \frac{\dot{I}_{\text{erase}}}{\epsilon_H}\right) H_{\text{obs}}(t), \qquad \frac{d\Psi}{dt} \geq -\mu_\Psi \Psi(t), \quad (10)$$

which by Grönwall's lemma guarantee strict positivity for all finite $t \geq 0$*:*

$$\gamma(t) \geq \gamma(0)e^{-bt} > 0, \qquad H_{\text{obs}}(t) \geq H_{\text{obs}}(0)e^{-(c\gamma_{\max} + \dot{I}_{\text{erase}}/\epsilon_H)t} > 0, \qquad \Psi(t) \geq \Psi(0)e^{-\mu_\Psi t} > 0. \quad (11)$$

Furthermore, when $\mathcal{P}_0 > 0$, *since* $0 \le \tanh(\dot{I}/\epsilon_I) \le \dot{I}/\epsilon_I$, $dH_{\rm obs}/dt \ge -\dot{I}_{\rm erase} \ge -M_H$, *and* $\mathcal{P}_{\rm eff} \ge \mathcal{P}_0$ *on* $\mathcal{B}_4$, *the processing rate obeys the uniform linear differential lower bound (*`dIdot_dt_4D_ge_linear_floor`*):*

$$\frac{d\dot{I}}{dt} \ge f\mathcal{P}_0 - \left(g + \frac{eM_H}{\epsilon_I}\right)\dot{I}(t), \tag{12}$$

which guarantees uniform asymptotic persistence away from the floor:

$$\begin{aligned} \liminf_{t\to\infty} \dot{I}(t) &\ge \dot{I}_{\rm floor}(\mathcal{P}_0) := \frac{f\mathcal{P}_0}{g + eM_H/\epsilon_I} > 0, \\ \liminf_{t\to\infty} \gamma(t) &\ge \gamma_{\rm floor}(\mathcal{P}_0) := \frac{a\mathcal{E}_{\rm avail}\dot{I}_{\rm floor}(\mathcal{P}_0)\gamma_{\max}}{b\gamma_{\max} + a\mathcal{E}_{\rm avail}\dot{I}_{\rm floor}(\mathcal{P}_0)} > 0. \end{aligned} \tag{13}$$

Proof. The 17 pointwise vector-field boundary and linear-comparison inequalities on $\mathcal{B}_4$—comprising all eight lower and upper faces, the entropy-rate bounds $-M_H \le dH_{\rm obs}/dt \le M_H$ (`dHobs_dt_4D_ge_neg_M_H` and `dHobs_dt_4D_le_M_H`), the flux bound $\mathcal{P}_{\rm eff} \le \mathcal{P}_{\rm eff,max}$ (`P_eff_le_P_eff_max`), and the linear comparison bounds $d\gamma/dt \ge -b\gamma$ (`dgamma_dt_4D_ge_neg_decay`) and $d\dot{I}/dt \ge f\mathcal{P}_0 - (g + eM_H/\epsilon_I)\dot{I}$ (`dIdot_dt_4D_ge_linear_floor`)—are machine-checked in `ExistenceVerification.lean` (zero `sorry` axioms, zero `nlinarith` tactics), from which continuous-time trajectory invariance $x(t) \in \mathcal{B}_4$ and strict interior persistence follow by classical application of the Nagumo–Bony–Brezis subtangentiality theorem and Grönwall's comparison lemma:

- **Faces 1 & 2 ($\gamma = 0$ and $\gamma = \gamma_{\max}$):** At $\gamma = 0$ with $\dot{I} \ge 0$, $d\gamma/dt = a\dot{I}\mathcal{E}_{\rm avail} \ge 0$ (`dgamma_dt_4D_nonneg_at_boundary`). At $\gamma = \gamma_{\max}$, $1 - \gamma_{\max}/\gamma_{\max} = 0$, so $d\gamma/dt = -b\gamma_{\max} < 0$ (`dgamma_dt_4D_neg_at_ceiling`).
- **Faces 3 & 4 ($H_{\rm obs} = 0$ and $H_{\rm obs} = H_{\max}$):** At $H_{\rm obs} = 0$, $\tanh(0/\epsilon_H) = 0$, so $dH_{\rm obs}/dt = c\gamma H_{\max} \ge 0$ (`dHobs_dt_4D_nonneg_at_boundary`). At $H_{\rm obs} = H_{\max}$, $dH_{\rm obs}/dt = -\dot{I}_{\rm erase}\tanh(H_{\max}/\epsilon_H) < 0$ (`dHobs_dt_4D_neg_at_ceiling`).
- **Faces 5 & 6 ($\dot{I} = 0$ and $\dot{I} = \dot{I}_{\max,4D}$):** At $\dot{I} = 0$, $\tanh(0/\epsilon_I) = 0$, so $d\dot{I}/dt = f(\mathcal{P}_0 + \lambda_{\rm sim} r_{\rm LRSC}\gamma + \lambda_{\rm lucid}\Psi)$, which is ≥ 0 for all $\mathcal{P}_0 \ge 0$ (`dIdot_dt_4D_nonneg_at_boundary_zero_drive`) and strictly > 0 for $\mathcal{P}_0 > 0$ (`dIdot_dt_4D_pos_at_boundary`). On $\mathcal{B}_4$, $0 \le \gamma \le \gamma_{\max}$ and $0 \le H_{\rm obs}$ imply $dH_{\rm obs}/dt \le c\gamma_{\max}H_{\max} \le M_H$ (`dHobs_dt_4D_le_M_H`), and $\gamma \le \gamma_{\max}, \Psi \le 1$ imply $\mathcal{P}_{\rm eff} \le \mathcal{P}_{\rm eff,max}$ (`P_eff_le_P_eff_max`). Chaining these bounds inside `dIdot_dt_4D_nonpos_at_ceiling` yields $d\dot{I}/dt\big|_{\dot{I}=\dot{I}_{\max,4D}} \le eM_H + f\mathcal{P}_{\rm eff,max} - g\dot{I}_{\max,4D} = 0$.
- **Faces 7 & 8 ($\Psi = 0$ and $\Psi = 1$):** By `R_self_sq_nonneg`, the state-dependent ratio $R_{\rm self}(s) = \mathrm{fb}(s)^2/(\mathrm{fb}(s)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ satisfies $R_{\rm self}(s) \ge 0$ for every $s \in \mathbb{R}^4$. Hence at $\Psi = 0$, $d\Psi/dt = \kappa_{\rm meta}R_{\rm self}(s) \ge 0$ (`dPsi_dt_4D_nonneg_at_boundary`), and at $\Psi = 1$, $d\Psi/dt = -\mu_\Psi < 0$ (`dPsi_dt_4D_neg_at_ceiling`).

For Part 2, since $0 \le \gamma(t) \le \gamma_{\max}$ and $\dot{I}(t) \ge 0$ on $\mathcal{B}_4$, the production term $a\dot{I}\mathcal{E}_{\rm avail}(1 - \gamma/\gamma_{\max}) \ge 0$, which yields $d\gamma/dt \ge -b\gamma$ (`dgamma_dt_4D_ge_neg_decay`). Similarly, for $H_{\rm obs} \ge 0$, $\tanh(H_{\rm obs}/\epsilon_H) \le H_{\rm obs}/\epsilon_H$, so $dH_{\rm obs}/dt = c\gamma H_{\max} - c\gamma H_{\rm obs} - \dot{I}_{\rm erase}\tanh(H_{\rm obs}/\epsilon_H) \ge -(c\gamma_{\max} + \dot{I}_{\rm erase}/\epsilon_H)H_{\rm obs}$. For $\Psi \in [0,1]$, $\kappa_{\rm meta}R_{\rm self}(1 - \Psi) \ge 0$ implies $d\Psi/dt \ge -\mu_\Psi\Psi$. Integrating these three linear differential inequalities gives Eq. (11). Finally, on $\mathcal{B}_4$, $c\gamma(H_{\max} - H_{\rm obs}) \ge 0$ and $\tanh(H_{\rm obs}/\epsilon_H) \le 1$ imply $dH_{\rm obs}/dt \ge -\dot{I}_{\rm erase} \ge -M_H$ (`dHobs_dt_4D_ge_neg_M_H`). Combining $0 \le \tanh(\dot{I}/\epsilon_I) \le \dot{I}/\epsilon_I$ with $dH_{\rm obs}/dt \ge -M_H$ and $\mathcal{P}_{\rm eff} \ge \mathcal{P}_0$ yields Eq. (12) (`dIdot_dt_4D_ge_linear_floor`). By standard linear ODE comparison, Eq. (12) forces $\liminf_{t\to\infty}\dot{I}(t) \ge \dot{I}_{\rm floor}(\mathcal{P}_0) > 0$, and substituting $\dot{I}(t) \ge \dot{I}_{\rm floor}(\mathcal{P}_0) - \varepsilon$ into Eq. (4) forces $\liminf_{t\to\infty}\gamma(t) \ge \gamma_{\rm floor}(\mathcal{P}_0) > 0$. □

Remark 4 (Exact Scope of the Lean 4 Formalization, Classical ODE Step, and Box Ceiling $\dot{I}_{\max,4D}$)**.** To keep the formal claims completely transparent, three points should be noted:

1. **Pointwise Vector-Field Inequalities in Lean 4 vs. Classical Nagumo/Grönwall Step:** `ExistenceVerification.lean` formalizes the algebraic vector field $F_{\text{exist}}(s)$ over Mathlib's $\mathbb{R}$ and machine-checks the 17 pointwise sign, ceiling, and linear-comparison inequalities on the boundary faces and interior of $\mathcal{B}_4$: `R_self_sq_nonneg` (proving non-negativity of the state-dependent function `R_self_sq` from `sq_nonneg`, `add_nonneg`, and `div_nonneg`), the derived ceiling bounds `dHobs_dt_4D_le_M_H` and `P_eff_le_P_eff_max`, the eight face theorems, and the linear decay comparison `dgamma_dt_4D_ge_neg_decay` ($d\gamma/dt \geq -b\gamma$). The unified theorem `existence_box_8face_nagumo_certificate` bundles the $\mathcal{P}_0 > 0$ case (where Face 5 is strictly positive, $d\dot{I}/dt|_{\dot{I}=0} > 0$), while the separate theorem `dIdot_dt_4D_nonneg_at_boundary_zero_drive` certifies Face 5 non-negativity ($d\dot{I}/dt|_{\dot{I}=0} \geq 0$) for all $\mathcal{P}_0 \geq 0$ (including $\mathcal{P}_0 = 0$). Passing from these machine-checked pointwise inequalities to continuous-time ODE trajectory invariance $x(t) \in \mathcal{B}_4$ and $\gamma(t) \geq \gamma(0)e^{-bt} > 0$ rests on the classical Nagumo–Bony–Brezis subtangentiality theorem and Grönwall's lemma applied to the smooth (C^∞) vector field.

2. **Offline Python Source Gate vs. Toolchain Compilation (`lake build`):** Layer 6 of `verification_existence_v1_0.py` performs a fast offline static audit (verifying comment-stripped absence of `sorry` and `nlinarith`, presence of all 17 theorem declarations and `#print axioms`, and `lakefile.lean` target registration) so that readers without a local Lean 4 installation can run the Python verification script standalone. Full Lean 4 type-checking and compilation of both `BoundaryVerification` and `ExistenceVerification` is executed via `lake build` under `leanprover/lean4:v4.36.0-rc1` (Mathlib `10156080d0e7`) and automated in `verify_ci.yml` (GitHub Actions CI).

3. **Conservative Analytical Ceiling $\dot{I}_{\max,4D}$ vs. Dynamical Trajectory Peak:** The analytical ceiling $\dot{I}_{\max,4D} = (eM_H + f\mathcal{P}_{\text{eff,max}})/g$ (which equals 3644.1 in Regime IVa) is an a priori worst-case Nagumo bound obtained by bounding $\tanh(\cdot) \leq 1$ and $\dot{H}_{\text{obs}} \leq M_H = c\gamma_{\max}H_{\max} + \dot{I}_{\text{erase}} = 1010$. Along actual post-Hopf relaxation orbits, $\gamma(t)$ and $H_{\max} - H_{\text{obs}}(t)$ never peak simultaneously, so the observed trajectory maximum is roughly an order of magnitude smaller ($\max_t \dot{I}(t) \approx 268.4$).

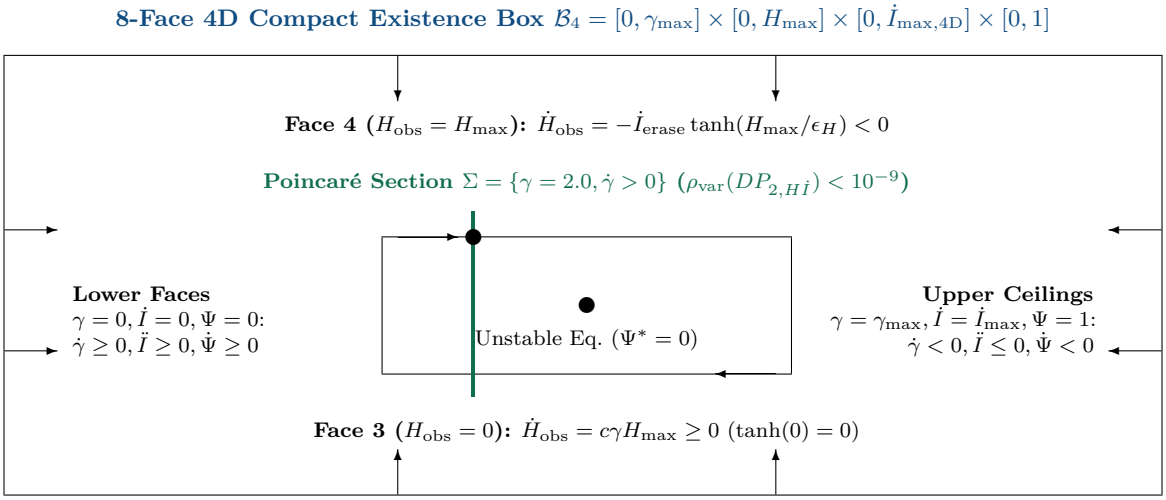


Figure 3: Geometric cross-section of the 8-face compact trapping box $\mathcal{B}_4$. All eight boundary and ceiling faces point inward (Theorem 3). When $e > e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}})$, the interior equilibrium $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*, \Psi^* = 0)$ is linearly unstable (Theorem 5), and trajectories in the interior basin converge numerically onto an attracting periodic orbit intersecting $\Sigma = \{\gamma = 2.0, \dot{\gamma} > 0\}$.

4 Global Hopf Monotonicity, Closed-Loop 4D Spectral Analysis & Zero-Drive Origin Stability

When $\lambda_{\text{sim}} = \lambda_{\text{lucid}} = 0$, the 3D $(\gamma, H_{\text{obs}}, \dot{I})$ subsystem runs on its own and simply drives the readout $\Psi(t)$. Once we turn on $\lambda_{\text{lucid}} > 0$ or $\lambda_{\text{sim}} > 0$, however, the four variables form a closed loop. To understand the stability of this 4D loop completely, we need to establish four things: (1) how λ_{lucid} and $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}}$ enter the 4×4 equilibrium Jacobian J_4; (2) the exact closed-loop Routh–Hurwitz tipping point $e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}})$; (3) the exact domain ($\mathcal{P}_0 > 10/99 \approx 0.1010$ in the $\epsilon_H, \epsilon_I \to 0^+$ limit, and $\mathcal{P}_0 \gtrsim 0.105$ at $\epsilon_H = \epsilon_I = 0.05$) where the open-loop rational formula $e_c(\mathcal{P}_0)$ applies and why it increases strictly with $\mathcal{P}_0$; and (4) the exact eigenvalues and local stability of the silent origin $(0, 0, 0, 0)$ when external drive is shut off completely ($\mathcal{P}_0 = 0$).

Theorem 5 (Closed-Loop Hopf Locus, $\epsilon \to 0^+$ Rational Domain, λ_{lucid} Decoupling, and $\mathcal{P}_0 = 0$ Origin Stability — `ANALYTIC_THEOREM`)**.** *Consider the smooth 4D system Eqs. (4)–(7) with quadratic self-feedback ratio R_{self} in Eq. (3):*

1. ***Exact Equilibrium*** $\Psi^* = 0$ ***and Block-Upper-Triangular Decoupling of*** λ_{lucid}***:*** *At any steady state $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*, \Psi^*)$, the balance condition $dH_{\text{obs}}/dt|_* = 0$ implies $\text{fb}^* = 0$, and therefore $R^*_{\text{self}} = 0$ and $\Psi^* = 0$ identically. Because $R_{\text{self}} = \text{fb}^2/(\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ is quadratic in* fb*, its gradient vanishes identically at equilibrium:*

$$\nabla_{(\gamma, H_{\text{obs}}, \dot{I})} R_{\text{self}}\Big|_{\dot{H}_{\text{obs}}=0} = (0, 0, 0). \tag{14}$$

Consequently, for ***all*** $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}} \geq 0$ *and* ***all*** $\lambda_{\text{lucid}} \geq 0$*, the 4×4 equilibrium Jacobian J_4 is block-upper-triangular:*

$$J_4(\Lambda_{\text{eff}}, \lambda_{\text{lucid}}) = \begin{pmatrix} -\alpha & 0 & \beta & 0 \\ \delta_\epsilon & -\mu_\epsilon & 0 & 0 \\ e_{\text{eff}}\delta_\epsilon + f\Lambda_{\text{eff}} & -e_{\text{eff}}\mu_\epsilon & -g & f\lambda_{\text{lucid}} \\ 0 & 0 & 0 & -\mu_\Psi \end{pmatrix}, \tag{15}$$

where $\alpha = b + a\mathcal{E}_{\text{avail}}\dot{I}^*/\gamma_{\max}$, $\beta = a\mathcal{E}_{\text{avail}}(1 - \gamma^*/\gamma_{\max})$, $\delta_\epsilon = c(H_{\max} - H^*_{\text{obs}})$, $\mu_\epsilon = c\gamma^* + (\dot{I}_{\text{erase}}/\epsilon_H)\,\text{sech}^2(H^*_{\text{obs}}/\epsilon_H)$, and $e_{\text{eff}} = e\tanh(\dot{I}^*/\epsilon_I)$. The characteristic polynomial factors identically as:

$$\det(\lambda I_4 - J_4) = (\lambda + \mu_\Psi)\,\det(\lambda I_3 - J_{3\times 3}(\Lambda_{\text{eff}}))\,, \qquad \frac{\partial}{\partial \lambda_{\text{lucid}}} \det(\lambda I_4 - J_4) \equiv 0. \tag{16}$$

Thus $\lambda_4 = -\mu_\Psi < 0$ is an exact stable eigenvalue, and the feedback coupling λ_{lucid} drops out of the local equilibrium stability spectrum completely (affecting trajectories only after they leave equilibrium and generate $\dot{H}_{\text{obs}} \neq 0$).

2. ***Exact Soft-Wall Closed-Loop Equilibrium and Routh–Hurwitz Hopf Locus*** $e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}})$***:*** *For any $\mathcal{P}_0 > 0$ and $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}} \geq 0$, the steady-state reconstruction capacity $\gamma^* \in (0, \gamma_{\max})$ is the unique positive root of the quadratic equation:*

$$a\mathcal{E}_{\text{avail}} f\Lambda_{\text{eff}}(\gamma^*)^2 + [bg\gamma_{\max} + a\mathcal{E}_{\text{avail}} f(\mathcal{P}_0 - \Lambda_{\text{eff}}\gamma_{\max})]\,\gamma^* - a\mathcal{E}_{\text{avail}} f\mathcal{P}_0\gamma_{\max} = 0, \tag{17}$$

with $\dot{I}^* = f(\mathcal{P}_0 + \Lambda_{\text{eff}}\gamma^*)/g$ and unique positive soft-wall entropy root $H^*_{\text{obs,sw}} \in (0, H_{\max})$ solving $c\gamma^*(H_{\max} - H^*_{\text{obs,sw}}) = \dot{I}_{\text{erase}}\tanh(H^*_{\text{obs,sw}}/\epsilon_H)$. By the Routh–Hurwitz criterion applied to $J_{3\times 3}(\Lambda_{\text{eff}})$, the interior equilibrium loses linear stability via a pair of purely imaginary eigenvalues $\lambda_{1,2} = \pm i\omega_c$ at the exact closed-loop Hopf threshold:

$$e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}}) = \frac{(\alpha + g)(\alpha + \mu_\epsilon)(g + \mu_\epsilon) - \beta f\Lambda_{\text{eff}}(\alpha + g)}{\beta\,\delta_\epsilon\,(\alpha + \mu_\epsilon + g)\,\tanh(\dot{I}^*/\epsilon_I)}. \tag{18}$$

3. $\epsilon \to 0^+$ ***Closed-Form Rational Locus*** $e_c(\mathcal{P}_0)$***, Its Domain*** $\mathcal{P}_0 > 10/99$***, and Global Monotonicity*** $de_c/d\mathcal{P}_0 > 0$*: On the open-loop v1.1/v1.5 benchmark slice* ($a = b = c = f = g = \mathcal{E}_{\rm avail} = 1$, $\gamma_{\max} = 10$, $H_{\max} = 100$, $\dot{I}_{\rm erase} = 10$, $\Lambda_{\rm eff} = 0$*), the unregularized* ($\epsilon_H, \epsilon_I \to 0^+$*) interior equilibrium* $H^*_{\rm obs,v1.4} = 100 - 10/\gamma^* = 100 - (10 + \mathcal{P}_0)/\mathcal{P}_0$ *is strictly positive if and only if* $\gamma^* > \gamma_{\rm crit} = \dot{I}_{\rm erase}/(cH_{\max}) = 0.10$*, which defines the exact interior positivity domain:*

$$\mathcal{P}_0 > \mathcal{P}_{\rm crit} = \frac{10}{99} = 0.\overline{10} \approx 0.101010\ldots \tag{19}$$

In the $\epsilon_H, \epsilon_I \to 0^+$ *limit on* $\mathcal{P}_0 > 10/99$*, Eq. (18) reduces to the closed-form rational function:*

$$e_c(\mathcal{P}_0) = \frac{\mathcal{P}_0(\mathcal{P}_0 + 20)(11\mathcal{P}_0 + 10)(\mathcal{P}_0^2 + 120\mathcal{P}_0 + 100)}{100(\mathcal{P}_0 + 10)(\mathcal{P}_0^2 + 130\mathcal{P}_0 + 200)}, \tag{20}$$

whose derivative with respect to $\mathcal{P}_0$ *is a ratio of polynomials with strictly positive integer coefficients:*

$$\frac{de_c}{d\mathcal{P}_0} = \frac{N_7(\mathcal{P}_0)}{100\left(\mathcal{P}_0^3 + 140\mathcal{P}_0^2 + 1500\mathcal{P}_0 + 2000\right)^2} > 0, \tag{21}$$

where the degree-7 numerator polynomial $N_7(\mathcal{P}_0)$ *has exclusively positive coefficients:*

$$\begin{aligned} N_7(\mathcal{P}_0) = {} & 22\mathcal{P}_0^7 + 6170\mathcal{P}_0^6 + 500000\mathcal{P}_0^5 + 11084000\mathcal{P}_0^4 \\ & + 99060000\mathcal{P}_0^3 + 241100000\mathcal{P}_0^2 + 188000000\mathcal{P}_0 + 40000000 > 0. \end{aligned} \tag{22}$$

At finite gate widths $\epsilon_H = \epsilon_I = \epsilon = 0.05$ *on* $\mathcal{P}_0 > 10/99$*, Eq. (18) relates to Eq. (20) via the two-gate identity* $e_c^{\rm CL}(\mathcal{P}_0, 0) = \left[e_c(\mathcal{P}_0) + O\left(\epsilon_H^{-1}\operatorname{sech}^2(H^*_{\rm obs,sw}/\epsilon_H)\right)\right] / \tanh(\mathcal{P}_0/\epsilon_I)$*. Because* $H^*_{\rm obs,v1.4}(\mathcal{P}_0) = 100 - (10 + \mathcal{P}_0)/\mathcal{P}_0$ *rises rapidly above zero once* $\mathcal{P}_0 > 10/99$*, by* $\mathcal{P}_0 = 0.105$ *we already have* $H^*_{\rm obs,sw} = 3.7619 > 75\epsilon_H$*, so the entropy-wall perturbation* $\operatorname{sech}^2(H^*_{\rm obs,sw}/\epsilon_H) = 1.78 \times 10^{-65}$ *vanishes to machine precision and* $e_c^{\rm CL}(\mathcal{P}_0, 0)\tanh(\mathcal{P}_0/\epsilon_I)$ *agrees with* $e_c(\mathcal{P}_0)$ *to* 10^{-64} *(while the multiplicative factor* $1/\tanh(\mathcal{P}_0/\epsilon_I)$ *yields* $|e_c^{\rm CL} - e_c| = 3.74 \times 10^{-4}$ *at* $\mathcal{P}_0 = 0.105$*,* 1.78×10^{-5} *at* $\mathcal{P}_0 = 0.20$*, and* $< 10^{-16}$ *for* $\mathcal{P}_0 \geq 1.0$*). Conversely, for* $0 < \mathcal{P}_0 \leq 10/99$ *(such as the Minimum Corner* $\mathcal{P}_0 = 0.10$ *where* $H^*_{\rm obs,v1.4} = -1 < 0$*), the system resides inside the* $O(\epsilon_H)$ *boundary layer* ($H^*_{\rm obs,sw} = 0.1295$*) where Eq. (18) gives* $e_c^{\rm CL}(0.10, 0) = 0.099132$*. At fixed observer gain* $e = 18/5 = 3.60$*:*

$$e_c(5) = \frac{377}{84} \approx 4.48809524 > e = \frac{18}{5} = 3.60, \tag{23}$$

$$e_c(4) = \frac{12069}{4025} \approx 2.99850932 > e_c(1) = \frac{97461}{364100} \approx 0.26767646. \tag{24}$$

4. ***Zero-Drive*** ($\mathcal{P}_0 = 0$) ***Origin Stability, Transcritical Threshold*** $\Lambda_{\rm eff} = 1$***, and Bistability:*** *When external sensory drive vanishes* ($\mathcal{P}_0 = 0$*), the origin* $x^*_{\rm origin} = (0, 0, 0, 0)$ *is an exact boundary equilibrium of Eqs. (4)–(7) because* $\tanh(0/\epsilon_H) = \tanh(0/\epsilon_I) = 0$ *and* $R_{\rm self}(0, 0, 0, 0) = 0/\sigma_0^2 = 0$*. Evaluating the* 4×4 *Jacobian at the origin yields:*

$$J_4(0, 0, 0, 0) = \begin{pmatrix} -b & 0 & a\mathcal{E}_{\rm avail} & 0 \\ cH_{\max} & -\dot{I}_{\rm erase}/\epsilon_H & 0 & 0 \\ f\Lambda_{\rm eff} & 0 & -g & f\lambda_{\rm lucid} \\ 0 & 0 & 0 & -\mu_\Psi \end{pmatrix}, \tag{25}$$

with characteristic polynomial $\det(\lambda I_4 - J_4(0)) = (\lambda + \dot{I}_{\rm erase}/\epsilon_H)(\lambda + \mu_\Psi)\left[\lambda^2 + (b + g)\lambda + (bg - a\mathcal{E}_{\rm avail} f\Lambda_{\rm eff})\right]$ *and exact eigenvalues:*

$$\lambda_{1,3} = \frac{-(b + g) \pm \sqrt{(b - g)^2 + 4a\mathcal{E}_{\rm avail} f\Lambda_{\rm eff}}}{2}, \qquad \lambda_2 = -\frac{\dot{I}_{\rm erase}}{\epsilon_H} < 0, \qquad \lambda_4 = -\mu_\Psi < 0. \tag{26}$$

Consequently, for $0 \leq \Lambda_{\rm eff} < \frac{bg}{a\mathcal{E}_{\rm avail} f}$ *(which equals 1 on the benchmark slice, including Regime IVb where* $\lambda_{\rm sim} = 7.20 \implies \Lambda_{\rm eff} = 0.80 < 1$ *with eigenvalues* $\lambda_1 = -1 + \sqrt{0.80} \approx -0.105573 < 0$*,*

$\lambda_3 = -1 - \sqrt{0.80} \approx -1.894427 < 0$, $\lambda_2 = -200 < 0$, $\lambda_4 = -0.60 < 0$*), **the origin** $(0,0,0,0)$ **is locally asymptotically stable**. At* $\Lambda_{\text{eff}} = \frac{bg}{a\mathcal{E}_{\text{avail}} f} = 1$*, the origin undergoes a transcritical bifurcation (*$\lambda_1 = 0$*), and for* $\Lambda_{\text{eff}} > 1$ *the origin becomes an unstable saddle as the nontrivial equilibrium* $\gamma_+^* = \gamma_{\max}\left(1 - \frac{bg}{a\mathcal{E}_{\text{avail}} f \Lambda_{\text{eff}}}\right) > 0$ *enters* $\mathcal{B}_4$*. Thus, in Regime IVb (*$\mathcal{P}_0 = 0, \Lambda_{\text{eff}} = 0.80$*), the system is **bistable**: persistence of the autonomous oscillation depends on the initial state lying inside the interior periodic-orbit basin rather than the local attraction neighborhood of* $(0,0,0,0)$.

Proof. Verified symbolically with zero residual in Layers 1 and 2 of `verification_existence_v1_0.py` using `SymPy` and `fractions.Fraction`. In Part 1, differentiating $f_4 = \kappa_{\text{meta}}\text{fb}^2(1-\Psi)/[\text{fb}^2 + (f\mathcal{P}_0)^2 + \sigma_0^2] - \mu_\Psi \Psi$ with respect to $v \in \{\gamma, H_{\text{obs}}, \dot{I}\}$ produces a factor of $2\,\text{fb}\,(\partial\text{fb}/\partial v)$, which vanishes identically whenever $\dot{H}_{\text{obs}} = 0$ ($\text{fb} = 0$). Expanding $\det(\lambda I_4 - J_4)$ along the fourth row $[0,0,0,-\mu_\Psi]$ yields Eq. (16). In Part 2, solving the Routh–Hurwitz boundary $c_2 c_1 = c_0$ of $J_{3\times3}(\Lambda_{\text{eff}})$ for $e = e_{\text{eff}}/\tanh(\dot{I}^*/\epsilon_I)$ gives Eq. (18). In Part 3, substituting $\gamma^* = 10\mathcal{P}_0/(10+\mathcal{P}_0)$ into $H^*_{\text{obs,v1.4}} = 100 - 10/\gamma^* > 0$ yields $\mathcal{P}_0 > 10/99$, and symbolic differentiation confirms Eq. (21). In Part 4, substituting $\mathcal{P}_0 = 0$ and $(\gamma, H_{\text{obs}}, \dot{I}, \Psi) = (0,0,0,0)$ into the partial derivatives of F_{exist} in `SymPy` yields Eq. (25) and its factored characteristic polynomial Eq. (26). □

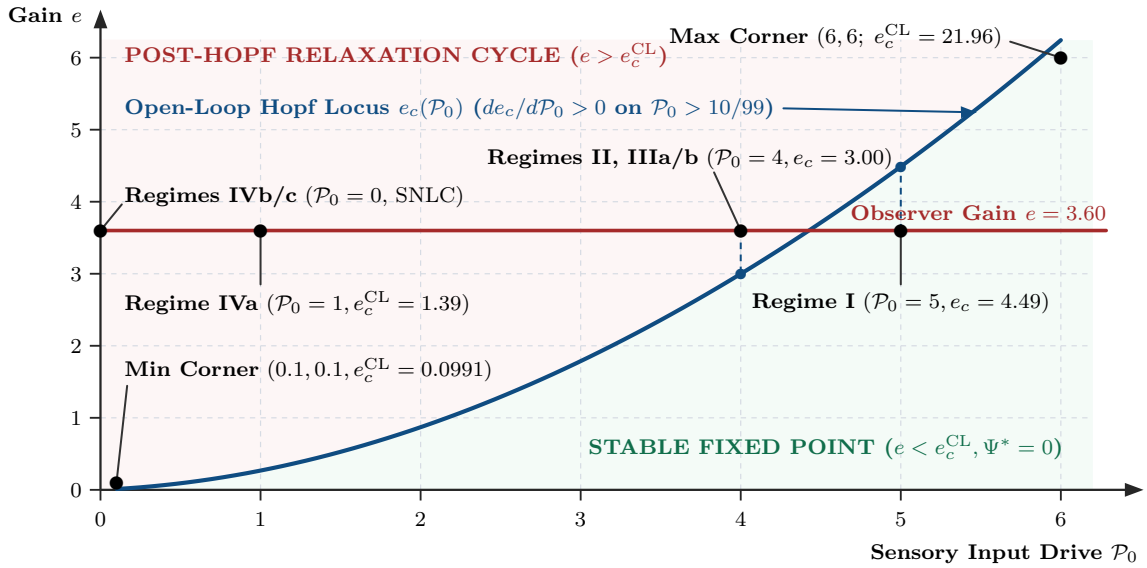


Figure 4: Bifurcation phase diagram in the $(\mathcal{P}_0, e)$ control plane showing the strictly increasing open-loop Hopf curve $e_c(\mathcal{P}_0)$ (Theorem 5), the fixed observer gain slice $e = 3.60$ spanning Regimes I–IVb, and the Minimum and Maximum parameter corners (at the Maximum corner, closed-loop feedback $\Lambda_{\text{eff}} = 10/9$ raises e_c^{CL} to $21.96 \gg 6$, stabilizing the fixed point).

5 Poincaré Integrals, 20D Variational Floquet Multipliers & the $\mathcal{P}_0 = 0$ Ψ-Memory SNLC Mechanism

When we measure the average value of a quantity along an oscillation, how we take that average matters. An adaptive solver like `LSODA` or `DOP853` automatically takes tiny steps around sharp turns (where $\dot{H}_{\text{obs}}$ and $\dot{I}$ change fast) and much bigger steps across slow, flat stretches. If we simply take the plain average of the numbers in the solver's array, the sharp corners get counted too many times, and changing the solver tolerance shifts the answer.

To obtain a step-size-invariant integral, Layer 3 of `verification_existence_v1_0.py` integrates two

exact accumulators alongside the 4D state variables:

$$\frac{dz_5}{dt} = \mathcal{P}_{\text{eff}}(\gamma(t), \Psi(t)), \qquad \frac{dz_6}{dt} = \operatorname{div} F_{\text{exist}}(x(t)) = \frac{\partial \dot{\gamma}}{\partial \gamma} + \frac{\partial \dot{H}_{\text{obs}}}{\partial H_{\text{obs}}} + \frac{\partial \ddot{I}}{\partial \dot{I}} + \frac{\partial \dot{\Psi}}{\partial \Psi}. \tag{27}$$

Between any two consecutive upward crossings $t_{k-1} < t_k$ of the Poincaré section $\Sigma = \{\gamma = 2.0, \dot{\gamma} > 0\}$, the oscillation period $T = t_k - t_{k-1}$, the **single-period time average** $\overline{\mathcal{P}}_{\text{eff},T}$, and the **Liouville orbital divergence integral** $\Delta_{\text{Liouville}}$ are:

$$\overline{\mathcal{P}}_{\text{eff},T} = \frac{1}{T}\int_{t_{k-1}}^{t_k} \mathcal{P}_{\text{eff}}(\gamma(t), \Psi(t))\, dt = \frac{z_5(t_k) - z_5(t_{k-1})}{t_k - t_{k-1}}, \qquad \Delta_{\text{Liouville}} = \int_{t_{k-1}}^{t_k} \operatorname{div} F_{\text{exist}}(x(t))\, dt. \tag{28}$$

To ensure every difference between regimes comes purely from the control parameters and not from hand-picked starting points, **all seven benchmark slices in Table 1 start from the exact same initial state** $x(0) = (2.05, 95.0, 4.0, 0.05)$ with unit-normalized dimensionless constants ($a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1$, $\gamma_{\max} = 10$, $H_{\max} = 100$, $\dot{I}_{\text{erase}} = 10$, $\epsilon_H = \epsilon_I = 0.05$, $\sigma_0 = 0.05$, $\mu_\Psi = 0.60$, $r_{\text{LRSC}} = 1/9$, $e = 3.60$).

Table 1: Summary of the seven dynamical benchmark slices inside $\mathcal{B}_4$ from common initial state $x(0) = (2.05, 95.0, 4.0, 0.05)$ at fixed gain $e = 3.60$ (rtol $= 10^{-10}$, atol $= 10^{-13}$). At $\mathcal{P}_0 = 0$ and $\Lambda_{\text{eff}} \le 1$ (Regimes IVb and IVc), no positive interior equilibrium or Hopf bifurcation exists (`N/A (SNLC)`); the periodic orbit is sustained by Ψ-memory above the saddle-node of limit cycles threshold $\lambda_{\text{lucid,SNLC}}$.

Regime Slice	$\mathcal{P}_0$	λ_{sim}	λ_{lucid}	κ_{meta}	e_c^{CL}	Period T	1-Period $\overline{\mathcal{P}}_{\text{eff},T}$	$[\min_t \Psi, \max_t \Psi]$	Poincaré Δ_Σ	Liouville $\int_0^T \operatorname{div} F\, dt$
I. High-Drive Fixed Pt	5.00	0.00	0.00	0.20	4.4881	Fixed Pt	5.000000	[0.0000, 0.0000]	5.62×10^{-10}	—
II. Low-Ψ Cycle	4.00	0.00	0.00	0.20	2.9985	18.828014	4.000000	[0.1650, 0.1902]	3.06×10^{-9}	−6929.8
IIIa. High-Ψ Readout	4.00	0.00	0.00	2.50	2.9985	18.828014	4.000000	[0.7069, 0.7913]	2.35×10^{-9}	−6948.1
IIIb. Coupled High-Ψ	4.00	0.00	0.30	2.50	2.9985	18.962844	4.180763	[0.7073, 0.7919]	3.13×10^{-9}	−6924.5
IVa. Low-Drive Coupled ($\mathcal{P}_0 = 1$)	1.00	7.20	0.30	2.50	1.3938	19.685223	3.266554	[0.7224, 0.7996]	2.60×10^{-9}	−7122.3
IVb. Zero-Drive Coupled ($\mathcal{P}_0 = 0$)	0.00	7.20	0.30	2.50	`N/A (SNLC)`	19.387103	2.190321	[0.7309, 0.8064]	2.68×10^{-9}	−7276.6
IVc. Pure Ψ-Memory ($\mathcal{P}_0 = 0$)	0.00	0.00	0.30	2.50	`N/A (SNLC)`	17.936755	0.239590	[0.7305, 0.8064]	1.74×10^{-9}	−7325.8

Three structural mechanisms emerge from Table 1, the 20D variational monodromy analysis (Table 3), and the zero-drive 2×2 ablation and continuation in λ_{lucid} (Table 4):

1. **Passive Readout Scaling (κ_{meta}) vs. Closed-Loop Modulation (λ_{lucid}):** Going from Regime II to Regime IIIa changes only the production weight κ_{meta} from 0.20 to 2.50 while keeping $\lambda_{\text{lucid}} = 0$. Because Ψ does not feed back into $\dot{I}$ when $\lambda_{\text{lucid}} = 0$, the 3D $(\gamma, H_{\text{obs}}, \dot{I})$ orbit in IIIa is identical to Regime II ($T = 18.828014$), while $\max \Psi(t)$ scales up from 0.1902 to 0.7913. Turning on $\lambda_{\text{lucid}} = 0.30$ in Regime IIIb closes the feedback loop: $\lambda_{\text{lucid}}\Psi(t)$ injects additional drive during the oscillation, lengthening the period by $+\Delta T = +0.134830$ to $T = 18.962844$ and raising $\overline{\mathcal{P}}_{\text{eff},T}$ to 4.180763, while leaving the equilibrium Hopf threshold $e_c^{\text{CL}} = 12069/4025 \approx 2.9985$ unchanged by Theorem 5(1).

2. **Why Ψ-Memory—Not the Instantaneous $\Lambda_{\text{eff}}\gamma$ Loop—Sustains the Zero-Drive Cycle at $\mathcal{P}_0 = 0$ (Table 4):** At $\mathcal{P}_0 = 0$ with $0 \le \Lambda_{\text{eff}} < 1$, substituting $\mathcal{P}_0 = 0$ into Eq. (17) yields $\Lambda_{\text{eff}}(\gamma^*)^2 + 10(1 - \Lambda_{\text{eff}})\gamma^* = 0$, whose roots are $\gamma^* = 0$ and $\gamma^* = -10(1 - \Lambda_{\text{eff}})/\Lambda_{\text{eff}} < 0$ (equal to $-2.5 < 0$ at $\Lambda_{\text{eff}} = 0.80$). Thus **no positive interior equilibrium or Hopf bifurcation exists at $\mathcal{P}_0 = 0$ for** $\Lambda_{\text{eff}} < 1$, and the origin $(0, 0, 0, 0)$ is locally asymptotically stable (Theorem 5(4)). A 2×2 ablation of $(\lambda_{\text{sim}}, \lambda_{\text{lucid}})$ at $\mathcal{P}_0 = 0$ (Table 4) reveals the exact physical mechanism that sustains the interior periodic orbit:

   - *Memoryless Proportional Loop Alone Collapses ($\lambda_{\text{sim}} = 7.20, \lambda_{\text{lucid}} = 0$):* During the deep erasure trough of the relaxation oscillation, $\gamma(t)$ drops to $\gamma_{\min} \approx 3.6 \times 10^{-4}$ and $\dot{I}(t)$ drops near zero. Because $\Lambda_{\text{eff}}\gamma(t)$ is purely proportional to the current state $\gamma(t)$, its feedback drive vanishes right at the trough ($\Lambda_{\text{eff}}\gamma_{\min} \approx 2.9 \times 10^{-4}$), allowing the locally stable origin to capture the trajectory ($\|x(400)\|_2 = 1.84 \times 10^{-16} < 10^{-12}$).

- *Slow Ψ-Memory Reservoir Bridges the Trough ($\lambda_{\rm sim}=0, \lambda_{\rm lucid}=0.30$, Regime IVc):* Because $\Psi(t)$ decays slowly at rate $\mu_\Psi = 0.60$ and is pumped toward 1 whenever $|\mathrm{fb}| \gg \sigma_0 = 0.05$, $\Psi(t)$ never drops below $\Psi_{\min} = 0.7305$ across the brief $\gamma \approx 0$ trough. The stored memory drive $\lambda_{\rm lucid}\Psi(t) \geq 0.219 > 0$ acts as a capacitive buffer that keeps $d\dot{I}/dt > 0$ at the trough and sustains the periodic cycle ($T = 17.936755$, $\max \dot{I} = 265.85$, $\overline{\mathcal{P}}_{\mathrm{eff},T} = 0.239590$).
- *Saddle-Node of Limit Cycles (SNLC) Continuation & Synergistic Flux Amplification:* Bisecting $\lambda_{\rm lucid}$ at $\mathcal{P}_0 = 0$ shows that the zero-drive cycle is born via a **saddle-node bifurcation of limit cycles (SNLC)** at $\lambda_{\mathrm{lucid,SNLC}}(0) = 0.080341$ when $\Lambda_{\rm eff} = 0$. When the proportional loop $\Lambda_{\rm eff} = 0.80$ ($\lambda_{\rm sim} = 7.20$, Regime IVb) is coupled alongside $\lambda_{\rm lucid}\Psi$, $\Lambda_{\rm eff}\gamma(t)$ amplifies the high-γ recovery stroke, increasing the period-averaged flux by 9.14× (from $\overline{\mathcal{P}}_{\mathrm{eff},T} = 0.239590$ to 2.190321) and lowering the critical SNLC threshold by 27.8% to $\lambda_{\mathrm{lucid,SNLC}}(0.80) = 0.057974$.

3. **20D First-Variation Monodromy Matrix $\Phi(T)$ & Saltation-Projected Transverse Floquet Multipliers (Table 3):** Rather than relying solely on finite-difference perturbations of the return map, Layer 4 of `verification_existence_v1_0.py` integrates the 20-dimensional first-variation monodromy system over one full period T from $x_\Sigma \in \Sigma = \{\gamma = 2.0, \dot{\gamma} > 0\}$:

$$\frac{dx}{dt} = F_{\rm exist}(x), \qquad \frac{d\Phi}{dt} = J_4(x(t))\,\Phi(t), \qquad \Phi(0) = I_4, \tag{29}$$

where $J_4(x) = \partial F_{\rm exist}/\partial x$ is verified against `SymPy` to 1.65×10^{-17}. At $t = T$, the monodromy matrix $M = \Phi(T)$ satisfies the periodic-orbit tangent eigenvector identity $\|MF_{\rm exist}(x_\Sigma) - F_{\rm exist}(x_\Sigma)\|_2 / \|F_{\rm exist}(x_\Sigma)\|_2 \leq 1.72 \times 10^{-9}$ (confirming the trivial tangent Floquet multiplier $\mu_{\rm tangent} = 1$). Projecting M onto the tangent space of the Poincaré section Σ (normal $n = (1,0,0,0)^\top$) via the exact saltation/Poincaré projection matrix:

$$P_\Sigma = I_4 - \frac{F_{\rm exist}(x(T))\, n^\top}{n^\top F_{\rm exist}(x(T))}, \qquad DP_{\rm var} = \big(P_\Sigma\, \Phi(T)\big)_{1:4,1:4}, \tag{30}$$

yields the 3×3 transverse return-map Jacobian $DP_{\rm var}$ directly from the variational equations. Because strong dissipation during the soft-wall boundary-layer stroke ($\int_0^T \mathrm{div}\, F_{\rm exist}\, dt \approx -6930 \implies \det M \approx 10^{-3010} \ll 1$) contracts the $(H_{\rm obs}, \dot{I})$ plane below IEEE-754 double-precision resolution ($\sim 10^{-12}$), the computed variational spectral radii $\rho_{\rm var}(DP_{2,H\dot{I}}) \approx 5.0 \times 10^{-12}$ to 6.6×10^{-12} represent strong double-precision numerical evidence of orbital contraction (whereas a formal computer-assisted proof would require interval-arithmetic ODE enclosures such as CAPD). In Regime II ($\lambda_{\rm lucid} = 0, \kappa_{\rm meta} = 0.20$), Ψ decouples from $(\gamma, H_{\rm obs}, \dot{I})$ and $(DP_{\rm var})_{3,3}$ matches the analytical orbital integral:

$$\mu_{\Psi,\rm exact} = \exp\left(-\int_0^T \left(\kappa_{\rm meta} R_{\rm self}(t) + \mu_\Psi\right) dt\right) = 2.537698 \times 10^{-6} \tag{31}$$

to $< 10^{-11}$.

6 Testing the Opposite Corners of the Parameter Cube

A good way to test whether a model truly holds together is to push all of its control knobs to their lowest and highest extremes at the same time. Across the five-parameter control cube $(e, \mathcal{P}_0, \kappa_{\rm meta}, \lambda_{\rm lucid}, \lambda_{\rm sim}) \in [0.10, 6.00] \times [0.00, 6.00] \times [0.05, 4.00] \times [0.00, 1.00] \times [0.00, 10.00]$, we examine both opposite corners:

1. **The Minimum Parameter Corner:** ($e = 0.10$, $\mathcal{P}_0 = 0.10$, $\kappa_{\rm meta} = 0.05$, $\lambda_{\rm lucid} = 0.00$, $\lambda_{\rm sim} = 0.00$), where sensory drive $\mathcal{P}_0 = 0.10 < \mathcal{P}_{\rm crit} = 10/99$ sits just below the unregularized positivity cutoff $\gamma_{\rm crit} = \dot{I}_{\rm erase}/(cH_{\max}) = 0.10$.
2. **The Maximum Parameter Corner:** ($e = 6.00$, $\mathcal{P}_0 = 6.00$, $\kappa_{\rm meta} = 4.00$, $\lambda_{\rm lucid} = 1.00$, $\lambda_{\rm sim} = 10.00$), where all five feedback and drive controls are turned up to their maximum values.

Table 2: Tolerance-convergence audit across rtol $\in \{10^{-8}, 10^{-10}, 10^{-12}\}$ (atol $\in \{10^{-11}, 10^{-13}, 10^{-14}\}$) for Regimes IIIb, IVa, and IVb, showing 6-digit convergence of the single-period integral $\overline{\mathcal{P}}_{\mathrm{eff},T}$ and oscillation period T, cross-solver agreement Δ_Σ (`LSODA` vs. `DOP853`), and 20D variational Floquet contraction.

Regime	Solver Tolerance	Period T	$\lvert T_{\mathrm{LS}} - T_{\mathrm{DP}}\rvert$	1-Period $\overline{\mathcal{P}}_{\mathrm{eff},T}$	$\lvert\Delta\overline{\mathcal{P}}_{\mathrm{eff},T}\rvert$	Poincaré Δ_Σ	Observed Multi-IC / Variational Floquet Bound
IIIb ($\mathcal{P}_0 = 4, \lambda_{\mathrm{luc}} = 0.3$)	rtol $= 10^{-8}$	18.962844	$< 5 \times 10^{-7}$	4.180763	$< 10^{-8}$	$< 5 \times 10^{-7}$	3-IC Spread $\Delta_{\mathrm{IC}} = 4.08 \times 10^{-11}$
IIIb ($\mathcal{P}_0 = 4, \lambda_{\mathrm{luc}} = 0.3$)	rtol $= 10^{-10}$	18.962844	$< 5 \times 10^{-8}$	4.180763	$< 10^{-9}$	$< 5 \times 10^{-8}$	Variational $\rho_{\mathrm{var}}(DP_3) = 1.87 \times 10^{-11}$
IIIb ($\mathcal{P}_0 = 4, \lambda_{\mathrm{luc}} = 0.3$)	rtol $= 10^{-12}$	18.962844	$< 5 \times 10^{-10}$	4.180763	$< 10^{-11}$	$< 5 \times 10^{-10}$	Liouville $\exp(\int_0^T \mathrm{div}\, F\, dt) \approx 10^{-3007} \ll 1$
IVa ($\mathcal{P}_0 = 1, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-8}$	19.685224	$< 5 \times 10^{-7}$	3.266554	$< 5 \times 10^{-8}$	$< 5 \times 10^{-7}$	3-IC Spread $\Delta_{\mathrm{IC}} = 2.58 \times 10^{-11}$
IVa ($\mathcal{P}_0 = 1, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-10}$	19.685223	$< 5 \times 10^{-8}$	3.266554	$< 10^{-9}$	$< 5 \times 10^{-8}$	Variational $\rho_{\mathrm{var}}(DP_3) = 4.96 \times 10^{-12}$
IVa ($\mathcal{P}_0 = 1, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-12}$	19.685223	$< 5 \times 10^{-10}$	3.266554	$< 10^{-10}$	$< 5 \times 10^{-10}$	Liouville $\exp(\int_0^T \mathrm{div}\, F\, dt) \approx 10^{-3093} \ll 1$
IVb ($\mathcal{P}_0 = 0, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-8}$	19.387103	$< 10^{-6}$	2.190321	$< 10^{-7}$	$< 5 \times 10^{-7}$	3-IC Spread $\Delta_{\mathrm{IC}} = 3.66 \times 10^{-11}$
IVb ($\mathcal{P}_0 = 0, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-10}$	19.387103	$< 5 \times 10^{-8}$	2.190321	$< 10^{-9}$	$< 5 \times 10^{-8}$	Variational $\rho_{\mathrm{var}}(DP_3) = 5.52 \times 10^{-12}$
IVb ($\mathcal{P}_0 = 0, \lambda_{\mathrm{sim}} = 7.2$)	rtol $= 10^{-12}$	19.387103	$< 5 \times 10^{-10}$	2.190321	$< 10^{-10}$	$< 5 \times 10^{-10}$	Liouville $\exp(\int_0^T \mathrm{div}\, F\, dt) \approx 10^{-3160} \ll 1$

Table 3: 20D first-variation monodromy and saltation-projected transverse Floquet multiplier audit (rtol $= 10^{-12}$, atol $= 10^{-14}$, Layer 4 of `verification_existence_v1_0.py`), alongside 3-initial-condition convergence Δ_{IC} and the analytical integral $\mu_{\Psi,\mathrm{exact}} = \exp(-\int_0^T (\kappa_{\mathrm{meta}} R_{\mathrm{self}} + \mu_\Psi)\, dt)$. Here $\|\Phi(T)F(x_\Sigma) - F(x_\Sigma)\|_2 / \|F(x_\Sigma)\|_2$ verifies the unit tangent Floquet multiplier $\mu_{\mathrm{tangent}} = 1$.

Oscillatory Regime	3-IC Spread Δ_{IC}	Tangent Residual $\|\Phi(T)F - F\|/\|F\|$	Variational $\rho_{\mathrm{var}}(DP_{2,HI})$	Analytical $\mu_{\Psi,\mathrm{exact}}$	Variational $\rho_{\mathrm{var}}(DP_3)$	Liouville $\exp(\int_0^T \mathrm{div}\, F\, dt)$
II ($\mathcal{P}_0 = 4, \kappa_{\mathrm{meta}} = 0.20, \lambda_{\mathrm{luc}} = 0$)	4.01×10^{-11}	1.60×10^{-9}	6.20×10^{-12}	2.537698×10^{-6}	2.537698×10^{-6}	$\approx 10^{-3010} \ll 1$
IIIb ($\mathcal{P}_0 = 4, \kappa_{\mathrm{meta}} = 2.50, \lambda_{\mathrm{luc}} = 0.3$)	4.08×10^{-11}	1.72×10^{-9}	6.64×10^{-12}	1.188720×10^{-14}	1.87×10^{-11}	$\approx 10^{-3007} \ll 1$
IVa ($\mathcal{P}_0 = 1, \lambda_{\mathrm{sim}} = 7.20, \lambda_{\mathrm{luc}} = 0.3$)	2.58×10^{-11}	1.35×10^{-9}	5.03×10^{-12}	5.805548×10^{-19}	4.96×10^{-12}	$\approx 10^{-3093} \ll 1$
IVb ($\mathcal{P}_0 = 0, \lambda_{\mathrm{sim}} = 7.20, \lambda_{\mathrm{luc}} = 0.3$)	3.66×10^{-11}	1.32×10^{-9}	5.52×10^{-12}	4.458069×10^{-26}	5.52×10^{-12}	$\approx 10^{-3160} \ll 1$
IVc ($\mathcal{P}_0 = 0, \lambda_{\mathrm{sim}} = 0.00, \lambda_{\mathrm{luc}} = 0.3$)	6.23×10^{-11}	9.05×10^{-10}	5.85×10^{-12}	5.451962×10^{-24}	5.85×10^{-12}	$\approx 10^{-3182} \ll 1$

Table 4: Zero-drive ($\mathcal{P}_0 = 0, e = 3.60, \kappa_{\mathrm{meta}} = 2.50$) 2×2 feedback ablation and Saddle-Node of Limit Cycles (SNLC) continuation in λ_{lucid} from $x(0) = (2.05, 95.0, 4.0, 0.05)$ (Layer 4 of `verification_existence_v1_0.py`).

Zero-Drive Configuration ($\mathcal{P}_0 = 0$)	$\Lambda_{\mathrm{eff}} = \lambda_{\mathrm{sim}}/9$	λ_{lucid}	Asymptotic Fate	Period T	1-Period $\overline{\mathcal{P}}_{\mathrm{eff},T}$	$[\min_t \Psi, \max_t \Psi]$	SNLC Threshold $\lambda_{\mathrm{lucid,SNLC}}$
1. Coupled Memory + Proportional (IVb)	0.80	0.30	Sustained Limit Cycle	19.387103	2.190321 (9.14×)	[0.7309, 0.8064]	$\lambda_{\mathrm{lucid,SNLC}}(0.80) = 0.057974$
2. Pure Ψ-Memory Loop Only (IVc)	0.00	0.30	Sustained Limit Cycle	17.936755	0.239590 (1.00×)	[0.7305, 0.8064]	$\lambda_{\mathrm{lucid,SNLC}}(0.00) = 0.080341$
3. Memoryless Proportional Only	0.80	0.00	Collapses to $(0,0,0,0)$	— ($\|x(400)\|_2 < 10^{-15}$)	0.000000	$\Psi(t) \to 0$	Fails ($\Lambda_{\mathrm{eff}} \gamma_{\min} \approx 3 \times 10^{-4} \to 0$)
4. Uncoupled Baseline	0.00	0.00	Collapses to $(0,0,0,0)$	— ($\|x(400)\|_2 < 10^{-20}$)	0.000000	$\Psi(t) \to 0$	—
5. Near-Origin Basin $x(0) = (10^{-3})^4$	0.80	0.30	Collapses to $(0,0,0,0)$	— ($\|x(320)\|_2 < 10^{-16}$)	0.000000	$\Psi(t) \to 0$	Bistable with Interior Orbit

MINIMUM PARAMETER CORNER

($e = 0.10,\ \mathcal{P}_0 = 0.10,\ \kappa_{\mathrm{meta}} = 0.05,\ \lambda_{\mathrm{luc}} = 0,\ \lambda_{\mathrm{sim}} = 0$)

- **Unregularized v1.4 Breakdown:** $\gamma^* = \frac{10}{101} \approx 0.09901 < 0.10 \implies H^*_{\mathrm{v1.4}} = 100 - 101 = -1.0 < 0$ ($\mathcal{P}_0 = 0.10 < 10/99$).
- **Soft-Wall Boundary-Layer Root:** $\tanh(H/\epsilon_H)$ rescues positivity at $H^*_{\mathrm{sw}} = +0.1295 > 0$ with $e_c^{\mathrm{CL}} = 0.099132 < e = 0.10$ ($e/e_c^{\mathrm{CL}} = 1.0088 > 1$).
- **Dahlquist RK4 Stiffness Bound:** Wall stiffness $|\lambda_{\max}| = \frac{e\dot{I}_{\mathrm{erase}}}{\epsilon_I} + g$ requires sub-stepping $\Delta t = 0.0005$ in the boundary layer ($T = 24.665558$).

MAXIMUM PARAMETER CORNER

($e = 6.00,\ \mathcal{P}_0 = 6.00,\ \kappa_{\mathrm{meta}} = 4.00,\ \lambda_{\mathrm{luc}} = 1.00,\ \lambda_{\mathrm{sim}} = 10.0$)

- **Exact Closed-Loop Quadratic:** $5(\gamma^*)^2 + 22\gamma^* - 270 = 0$ yields the exact algebraic steady state:

$$\gamma^* = \frac{-11 + \sqrt{1471}}{5} \approx 5.47072356,$$

$$\mathcal{P}^*_{\mathrm{eff}} = \frac{32 + 2\sqrt{1471}}{9} \approx 12.07858174.$$

- **Closed-Loop Stabilization:** Strong feedback $\Lambda_{\mathrm{eff}} = 10/9$ raises the closed-loop Hopf threshold to $e_c^{\mathrm{CL}} = 21.958110 \gg e = 6.00$ ($e/e_c^{\mathrm{CL}} = 0.2732 < 1$), locking the system to a stable fixed point ($\Psi^* \to 0$).

Figure 5: Analytical and numerical comparison of the 4D system at the Minimum vs. Maximum corners of the parameter cube, verified in Layer 5 of `verification_existence_v1_0.py`.

6.1 The Minimum Corner ($e = 0.10, \mathcal{P}_0 = 0.10$) and Why Soft Walls Need Fine Steps

At the Minimum corner ($e = 0.10, \mathcal{P}_0 = 0.10, \kappa_{\text{meta}} = 0.05, \lambda_{\text{lucid}} = 0, \lambda_{\text{sim}} = 0$), the steady processing rate is $\dot{I}^* = 0.10$ and the steady reconstruction capacity is:

$$\gamma^* = \frac{10(0.10)}{10 + 0.10} = \frac{1}{10.1} = \frac{10}{101} \approx 0.09900990. \tag{32}$$

In the old unregularized model, because $\mathcal{P}_0 = 0.10 < \mathcal{P}_{\text{crit}} = 10/99 \approx 0.10101$ ($\gamma^* = 10/101 < \gamma_{\text{crit}} = 0.10$), the unregularized formula gives $H^*_{\text{obs,v1.4}} = 100 - 10/(10/101) = -1.0 < 0$. That means no positive steady state even existed at $\mathcal{P}_0 = 0.10$ without the soft wall, which is why the $\epsilon \to 0^+$ polynomial Eq. (20) (0.011607) does not apply here.

In our soft-wall model, the gate $\tanh(H_{\text{obs}}/\epsilon_H)$ with $\epsilon_H = 0.05$ catches the entropy just above zero at a positive boundary-layer balance point $H^*_{\text{obs,sw}} \in (0, H_{\max})$:

$$c\gamma^*(H_{\max} - H^*_{\text{obs,sw}}) = \dot{I}_{\text{erase}} \tanh\left(\frac{H^*_{\text{obs,sw}}}{\epsilon_H}\right) \implies H^*_{\text{obs,sw}} = 0.12951822 > 0. \tag{33}$$

Evaluating the exact soft-wall threshold Eq. (18) at this boundary-layer root gives $e_c^{\text{CL}}(0.10, 0) = 0.09913224 < e = 0.10$ ($e/e_c^{\text{CL}} = 1.0088 > 1$). Because $e = 0.10$ sits 0.88% above the soft-wall threshold, the system enters a slow, gentle periodic cycle with period $T = 24.665558$, $\gamma(t) \in [0.0050, 2.3181]$, $H_{\text{obs}}(t) \in [0.0025, 94.6536]$, $\dot{I}_{\min} = 4.95 \times 10^{-3} > 0$, and $\max \Psi(t) = 0.0707$, with `LSODA` and `DOP853` agreeing on the Poincaré section to $9.04 \times 10^{-9} < 10^{-7}$ at rtol $= 10^{-10}$ ($8.60 \times 10^{-11} < 10^{-8}$ at rtol $= 10^{-12}$).

This corner also shows clearly why fixed-step Runge–Kutta (RK4) integrators need smaller steps near the walls:

- **Entropy Wall Stiffness ($H_{\text{obs}} \to 0^+$):** Near $H_{\text{obs}} \approx 0$, the slope $\partial\dot{H}_{\text{obs}}/\partial H_{\text{obs}} = -c\gamma - (\dot{I}_{\text{erase}}/\epsilon_H)\,\text{sech}^2(H_{\text{obs}}/\epsilon_H)$ reaches $-\dot{I}_{\text{erase}}/\epsilon_H = -10/0.05 = -200$. Since classical 4th-order RK4 is stable on $[-2.78529, 0]$, staying stable near $H_{\text{obs}} \approx 0$ requires $\Delta t < 2.78529/200 \approx 0.01393$.
- **Processing Wall Stiffness ($\dot{I} \to 0^+$):** Near $\dot{I} \approx 0$ when $\dot{H}_{\text{obs}} \approx -\dot{I}_{\text{erase}}$, the slope $\partial\ddot{I}/\partial\dot{I}$ reaches $-(e\dot{I}_{\text{erase}}/\epsilon_I + g)$, which is -721 at $e = 3.60$ ($\Delta t_{\text{crit},\dot{I}} = 0.00386$) and -1201 at $e = 6.00$ ($\Delta t_{\text{crit},\dot{I}} = 0.00232$). Sub-stepping RK4 at $\Delta t = 0.0005 \ll 0.00232$ whenever $\min(H_{\text{obs}}, \dot{I}) < 2.5$ keeps trajectories strictly inside $\mathcal{B}_4$ everywhere.

6.2 The Maximum Corner ($e = 6, \mathcal{P}_0 = 6, \lambda_{\text{sim}} = 10$) and Closed-Loop Stabilization

At the Maximum corner ($e = 6.00, \mathcal{P}_0 = 6.00, \kappa_{\text{meta}} = 4.00, \lambda_{\text{lucid}} = 1.00, \lambda_{\text{sim}} = 10.0$), the effective loop gain is $\Lambda_{\text{eff}} = \lambda_{\text{sim}} r_{\text{LRSC}} = 10/9$. Plugging $\mathcal{P}_0 = 6$ and $\Lambda_{\text{eff}} = 10/9$ into Eq. (17) gives:

$$\frac{10}{9}(\gamma^*)^2 + \left(10 + 6 - \frac{100}{9}\right)\gamma^* - 60 = 0 \iff 5(\gamma^*)^2 + 22\gamma^* - 270 = 0, \tag{34}$$

whose exact positive root and steady-state flux are:

$$\gamma^*_{\text{max-corner}} = \frac{-11 + \sqrt{1471}}{5} \approx 5.47072356, \qquad \mathcal{P}^*_{\text{eff}} = \dot{I}^* = 6 + \frac{10}{9}\gamma^* = \frac{32 + 2\sqrt{1471}}{9} \approx 12.07858174, \tag{35}$$

with $H^*_{\text{obs}} = 100 - 10/\gamma^* \approx 98.17208807$. At this high-capacity balance point, Theorem 5(2) gives the closed-loop threshold $e_c^{\text{CL}}(6.0, 10/9) = 21.95810954$. Even with $e = 6.00$ at its maximum setting, the system sits well below the tipping point ($e/e_c^{\text{CL}} = 0.273248 < 1$), so fluctuations damp out quickly and the state locks onto the steady point with $\Psi^* = 0$.

Table 5: Exact analytical and multi-solver (`LSODA`/`DOP853`, rtol $= 10^{-10}$ and rtol $= 10^{-12}$) comparison of the Minimum vs. Maximum corners of the 5-parameter control cube.

Parameter Corner	$(e, \mathcal{P}_0, \kappa_{\text{meta}}, \lambda_{\text{luc}}, \lambda_{\text{sim}})$	e_c^{CL}	e/e_c^{CL}	**Asymptotic** $\gamma(t)$	**Asymptotic** $H_{\text{obs}}(t)$	$\max_t \Psi(t)$	**1-Period / Steady** $\overline{\mathcal{P}}_{\text{eff}}$
Minimum Corner	$(0.10, 0.10, 0.05, 0.00, 0.00)$	0.099132	$1.0088 > 1$	$[0.0050, 2.3181]$	$[0.0025, 94.6536]$	0.0707	0.100000 ($T = 24.6656$)
Maximum Corner	$(6.00, 6.00, 4.00, 1.00, 10.0)$	21.958110	$0.2732 < 1$	5.470724 (Fixed Pt)	98.172088 (Fixed Pt)	0.0000 ($\Psi^* \to 0$)	12.078582 (Exact $\frac{32+2\sqrt{1471}}{9}$)

7 Complete Lean 4 Pointwise Boundary & Linear-Floor Module (`ExistenceVerification.lean`)

Below is the complete Lean 4 proof module `ExistenceVerification.lean` (17 theorems, registered in `lakefile.lean` as default library target `ExistenceVerification` alongside `BoundaryVerification`), proving unconditional non-negativity of the state-dependent quadratic ratio `R_self_sq`, the derived upper and lower entropy-rate bounds `dHobs_dt_4D_le_M_H` and `dHobs_dt_4D_ge_neg_M_H`, the pointwise linear comparison bounds `dgamma_dt_4D_ge_neg_decay` and `dIdot_dt_4D_ge_linear_floor`, the zero-drive Face 5 theorem `dIdot_dt_4D_nonneg_at_boundary_zero_drive`, the derived Face 6 ceiling bound `dIdot_dt_4D_nonpos_at_ceiling`, the unified 8-face certificate `existence_box_8face_nagumo_certificate`, and the `#print axioms` audit commands.

```
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Deriv
import Mathlib.Topology.MetricSpace.Basic
import Mathlib.Tactic

/-!
# Pointwise Boundary-Face & Linear-Comparison Algebra for the 4D Soft-Wall ODE System (`B_4`)
## Appendix B v1.0 --- Bounded Self-Referential Dynamics & Psi-Memory Persistence

This Lean 4 module (`ExistenceVerification.lean`, registered as a default library target in
`lakefile.lean` alongside `BoundaryVerification.lean` under `leanprover/lean4:v4.36.0-rc1`
and Mathlib commit `10156080d0e7`) formalizes the algebraic vector field
`F_exist = (dgamma_dt_4D, dHobs_dt_4D, dIdot_dt_4D, dPsi_dt_4D)` on the dimensionless 4D state
`ExistenceState = (gamma, H_obs, I_dot, Psi)` with the smooth quadratic self-feedback ratio:
`R_self_sq(s) = fb(s)^2 / (fb(s)^2 + (f * P_0)^2 + sigma_0^2)`.

Scope of this module (17 machine-checked pointwise algebraic theorems, **zero `sorry` axioms,
zero `nlinarith` tactics**):
1. **Unconditional Non-Negativity of State-Dependent `R_self_sq`** (`R_self_sq_nonneg`):
   Proves `0 <= R_self_sq ... s` from `sq_nonneg`, `add_nonneg`, and `div_nonneg`.
2. **Derived Upper & Lower Entropy-Rate Bounds on `B_4`** (`dHobs_dt_4D_le_M_H` and
   `dHobs_dt_4D_ge_neg_M_H`): Proves `-M_H_4D <= dHobs_dt_4D <= M_H_4D` directly from box
   membership (`0 <= s.gamma <= gamma_max`, `0 <= s.H_obs <= H_max`).
3. **All Eight Pointwise Boundary-Face Sign Conditions on `B_4`** (Faces 1--8, including both
   `P_0 > 0` in `dIdot_dt_4D_pos_at_boundary` and `P_0 >= 0` in
   `dIdot_dt_4D_nonneg_at_boundary_zero_drive`).
4. **Pointwise Linear Comparison Lower Bounds for `gamma` and `I_dot`** (`dgamma_dt_4D_ge_neg_decay`
   proving `dgamma/dt >= -b gamma`, and `dIdot_dt_4D_ge_linear_floor` proving
   `dI_dot/dt >= f P_0 - e ((I_dot / eps_I) M_H_4D) - g I_dot` on `B_4`). Continuous-time forward invariance
   and Gronwall / uniform `liminf` persistence follow from these pointwise inequalities via
   classical ODE comparison theorems in the accompanying manuscript.
5. **Unified 8-Face 4D Box Pointwise Certificate** (`existence_box_8face_nagumo_certificate`).
-/

noncomputable section

/-- Dimensionless 4D state-space coordinates `(gamma, H_obs, I_dot, Psi)` of the soft-wall ODE system. -/
structure ExistenceState where
  gamma : Real -- Dimensionless reconstruction capacity (gamma in [0, gamma_max])
  H_obs : Real -- Dimensionless internal entropy / uncertainty coordinate (H_obs in [0, H_max])
  I_dot : Real -- Dimensionless processing rate coordinate (I_dot in [0, I_dot_max_4D])
  Psi : Real -- Dimensionless slow-memory / self-feedback coordinate (Psi in [0, 1])

/-- Effective input drive:
`P_eff(gamma, Psi) = P_0 + lam_sim * r_lrsc * gamma + lam_lucid * Psi`.
Note that `lam_sim` and `r_lrsc` enter solely through their product `Lam_eff = lam_sim * r_lrsc`. -/
def P_eff (P_0 lam_sim r_lrsc lam_lucid : Real) (s : ExistenceState) : Real :=
  P_0 + lam_sim * r_lrsc * s.gamma + lam_lucid * s.Psi

/-- Upper bound `P_eff_max = P_0 + lam_sim * r_lrsc * gamma_max + lam_lucid` on `[0, gamma_max] x [0, 1]`. -/
```

```
def P_eff_max (P_0 lam_sim r_lrsc gamma_max lam_lucid : Real) : Real :=
  P_0 + lam_sim * r_lrsc * gamma_max + lam_lucid

/-- Eq. (B.1): Reconstruction capacity dynamics `dgamma/dt = a I_dot E_avail (1 - gamma / gamma_max) - b gamma`. -/
def dgamma_dt_4D (a b E_avail gamma_max : Real) (s : ExistenceState) : Real :=
  a * s.I_dot * E_avail * (1 - s.gamma / gamma_max) - b * s.gamma

/-- Eq. (B.2): Soft-wall entropy dynamics `dH_obs/dt = c gamma (H_max - H_obs) - I_dot_erase tanh(H_obs / eps_H)`.
    -/
def dHobs_dt_4D (c H_max I_erase epsH : Real) (s : ExistenceState) : Real :=
  c * s.gamma * (H_max - s.H_obs) - I_erase * Real.tanh (s.H_obs / epsH)

/-- Eq. (B.3): Coupled processing dynamics
`dI_dot/dt = e tanh(I_dot / eps_I) (dH_obs/dt) + f P_eff(gamma, Psi) - g I_dot`. -/
def dIdot_dt_4D (e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI : Real)
    (s : ExistenceState) : Real :=
  e * Real.tanh (s.I_dot / epsI) * (dHobs_dt_4D c H_max I_erase epsH s) +
    f * (P_eff P_0 lam_sim r_lrsc lam_lucid s) - g * s.I_dot

/-- Smooth (`C^infty`) quadratic self-feedback ratio:
`R_self_sq(s) = fb(s)^2 / (fb(s)^2 + (f * P_0)^2 + sigma_0^2)`, where
`fb(s) = e * tanh(I_dot / eps_I) * dH_obs/dt` and `sigma_0 > 0` guarantees smoothness for all `P_0 >= 0`. -/
def R_self_sq (e f P_0 sigma_0 c H_max I_erase epsH epsI : Real) (s : ExistenceState) : Real :=
  let fb := e * Real.tanh (s.I_dot / epsI) * (dHobs_dt_4D c H_max I_erase epsH s)
  (fb ^ 2) / (fb ^ 2 + (f * P_0) ^ 2 + sigma_0 ^ 2)

/-- Eq. (B.4): Slow-memory / self-feedback readout (`Psi`) dynamics with
the explicit state-dependent smooth quadratic ratio `R_self_sq`:
`dPsi/dt = kappa_meta * R_self_sq(s) * (1 - Psi) - mu_Psi * Psi`. -/
def dPsi_dt_4D (kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI : Real)
    (s : ExistenceState) : Real :=
  kappa_meta * (R_self_sq e f P_0 sigma_0 c H_max I_erase epsH epsI s) * (1 - s.Psi) - mu_Psi * s.Psi

/-- Maximum entropy-rate magnitude bound `M_H_4D = c gamma_max H_max + I_dot_erase`. -/
def M_H_4D (c gamma_max H_max I_erase : Real) : Real :=
  c * gamma_max * H_max + I_erase

/-- Upper processing ceiling `I_dot_max_4D = (e M_H_4D + f P_eff_max) / g` of the 4D box `B_4`. -/
def I_dot_max_4D (e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max I_erase : Real) : Real :=
  (e * M_H_4D c gamma_max H_max I_erase + f * P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid) / g

/-- Unified 4D vector field `F_exist : ExistenceState -> ExistenceState`. -/
def F_exist (a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
    I_erase epsH epsI kappa_meta mu_Psi : Real) (s : ExistenceState) : ExistenceState :=
  { gamma := dgamma_dt_4D a b E_avail gamma_max s
  , H_obs := dHobs_dt_4D c H_max I_erase epsH s
  , I_dot := dIdot_dt_4D e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI s
  , Psi := dPsi_dt_4D kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI s }

/-- Theorem B.0 (Unconditional Non-Negativity of State-Dependent `R_self_sq`):
For all real parameters and every state `s : ExistenceState`, `0 <= R_self_sq ... s`. -/
theorem R_self_sq_nonneg
    (e f P_0 sigma_0 c H_max I_erase epsH epsI : Real) (s : ExistenceState) :
    0 <= R_self_sq e f P_0 sigma_0 c H_max I_erase epsH epsI s := by
  unfold R_self_sq
  dsimp only
  have h_num : 0 <= (e * Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) ^ 2 :=
    sq_nonneg _
  have h_den1 : 0 <= (e * Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) ^ 2 + (f * P_0) ^ 2 :=
    add_nonneg h_num (sq_nonneg _)
  have h_den : 0 <= (e * Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) ^ 2 + (f * P_0) ^ 2 +
      sigma_0 ^ 2 :=
    add_nonneg h_den1 (sq_nonneg _)
  exact div_nonneg h_num h_den

/-- Theorem B.1 (Face 1 --- Lower gamma-Boundary Condition, gamma = 0): -/
theorem dgamma_dt_4D_nonneg_at_boundary
    (a b E_avail gamma_max : Real) (ha : 0 <= a) (hE : 0 <= E_avail)
    (s : ExistenceState) (hgamma : s.gamma = 0) (hI : 0 <= s.I_dot) :
    0 <= dgamma_dt_4D a b E_avail gamma_max s := by
  unfold dgamma_dt_4D
  rw [hgamma, zero_div, sub_zero, mul_one, mul_zero, sub_zero]
  exact mul_nonneg (mul_nonneg ha hI) hE

/-- Theorem B.1b (Pointwise Linear Decay Lower Bound for gamma on `B_4`):
For any state `s` in the box (`0 <= s.I_dot`, `s.gamma <= gamma_max`, `0 < gamma_max`), the production
```

```
/-- term `a * s.I_dot * E_avail * (1 - s.gamma / gamma_max)` is non-negative, so `dgamma_dt_4D >= -b * s.gamma`. -/
theorem dgamma_dt_4D_ge_neg_decay
    (a b E_avail gamma_max : Real)
    (ha : 0 <= a) (hE : 0 <= E_avail) (hgmax : 0 < gamma_max)
    (s : ExistenceState) (hI : 0 <= s.I_dot) (hgamma_le : s.gamma <= gamma_max) :
    -b * s.gamma <= dgamma_dt_4D a b E_avail gamma_max s := by
  unfold dgamma_dt_4D
  have h_div_le : s.gamma / gamma_max <= 1 := (div_le_one hgmax).mpr hgamma_le
  have h_one_sub : 0 <= 1 - s.gamma / gamma_max := sub_nonneg.mpr h_div_le
  have h_prod1 : 0 <= a * s.I_dot := mul_nonneg ha hI
  have h_prod2 : 0 <= a * s.I_dot * E_avail := mul_nonneg h_prod1 hE
  have h_prod3 : 0 <= a * s.I_dot * E_avail * (1 - s.gamma / gamma_max) :=
    mul_nonneg h_prod2 h_one_sub
  linarith

/-- Theorem B.2 (Face 2 --- Upper gamma-Ceiling Condition, gamma = gamma_max): -/
theorem dgamma_dt_4D_neg_at_ceiling
    (a b E_avail gamma_max : Real) (hb : 0 < b) (hgmax : 0 < gamma_max)
    (s : ExistenceState) (hgamma : s.gamma = gamma_max) :
    dgamma_dt_4D a b E_avail gamma_max s < 0 := by
  unfold dgamma_dt_4D
  have hdiv : gamma_max / gamma_max = 1 := div_self (ne_of_gt hgmax)
  rw [hgamma, hdiv, sub_self, mul_zero, zero_sub]
  exact neg_lt_zero.mpr (mul_pos hb hgmax)

/-- Theorem B.3 (Face 3 --- Lower H_obs-Boundary Condition, H_obs = 0): -/
theorem dHobs_dt_4D_nonneg_at_boundary
    (c H_max I_erase epsH : Real) (hc : 0 <= c) (hHmax : 0 <= H_max)
    (s : ExistenceState) (hH : s.H_obs = 0) (hgamma : 0 <= s.gamma) :
    0 <= dHobs_dt_4D c H_max I_erase epsH s := by
  unfold dHobs_dt_4D
  rw [hH, zero_div, Real.tanh_zero, mul_zero, sub_zero, sub_zero]
  exact mul_nonneg (mul_nonneg hc hgamma) hHmax

/-- Theorem B.4 (Face 4 --- Upper H_obs-Ceiling Condition, H_obs = H_max): -/
theorem dHobs_dt_4D_neg_at_ceiling
    (c H_max I_erase epsH : Real) (hIe : 0 < I_erase) (htanh : 0 < Real.tanh (H_max / epsH))
    (s : ExistenceState) (hH : s.H_obs = H_max) :
    dHobs_dt_4D c H_max I_erase epsH s < 0 := by
  unfold dHobs_dt_4D
  rw [hH, sub_self, mul_zero, zero_sub]
  exact neg_lt_zero.mpr (mul_pos hIe htanh)

/-- Lemma B.5a (Positivity of Effective Flux `P_eff` for `P_0 > 0`): -/
theorem P_eff_pos_on_box
    (P_0 lam_sim r_lrsc lam_lucid : Real)
    (hP0 : 0 < P_0) (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (s : ExistenceState) (hgamma : 0 <= s.gamma) (hPsi : 0 <= s.Psi) :
    0 < P_eff P_0 lam_sim r_lrsc lam_lucid s := by
  unfold P_eff
  have h1 : 0 <= lam_sim * r_lrsc := mul_nonneg hsim hr
  have h2 : 0 <= lam_sim * r_lrsc * s.gamma := mul_nonneg h1 hgamma
  have h3 : 0 <= lam_lucid * s.Psi := mul_nonneg hluc hPsi
  linarith

/-- Theorem B.5 (Face 5 --- Lower I_dot-Boundary Strict Positivity for `P_0 > 0`, I_dot = 0): -/
theorem dIdot_dt_4D_pos_at_boundary
    (e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI : Real)
    (hf : 0 < f) (hP0 : 0 < P_0) (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (s : ExistenceState) (hI : s.I_dot = 0) (hgamma : 0 <= s.gamma) (hPsi : 0 <= s.Psi) :
    0 < dIdot_dt_4D e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI s := by
  unfold dIdot_dt_4D
  rw [hI, zero_div, Real.tanh_zero, mul_zero, zero_mul, zero_add, mul_zero, sub_zero]
  have hPeff : 0 < P_eff P_0 lam_sim r_lrsc lam_lucid s :=
    P_eff_pos_on_box P_0 lam_sim r_lrsc lam_lucid hP0 hsim hr hluc s hgamma hPsi
  exact mul_pos hf hPeff

/-- Theorem B.5b (Face 5 Zero-Drive Extension --- Non-Negativity for `P_0 >= 0`, including `P_0 = 0`): -/
theorem dIdot_dt_4D_nonneg_at_boundary_zero_drive
    (e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI : Real)
    (hf : 0 <= f) (hP0 : 0 <= P_0) (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (s : ExistenceState) (hI : s.I_dot = 0) (hgamma : 0 <= s.gamma) (hPsi : 0 <= s.Psi) :
    0 <= dIdot_dt_4D e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI s := by
  unfold dIdot_dt_4D P_eff
  rw [hI, zero_div, Real.tanh_zero, mul_zero, zero_mul, zero_add, mul_zero, sub_zero]
  have h1 : 0 <= lam_sim * r_lrsc := mul_nonneg hsim hr
```

```
  have h2 : 0 <= lam_sim * r_lrsc * s.gamma := mul_nonneg h1 hgamma
  have h3 : 0 <= lam_lucid * s.Psi := mul_nonneg hluc hPsi
  have h4 : 0 <= P_0 + lam_sim * r_lrsc * s.gamma + lam_lucid * s.Psi := by linarith
  exact mul_nonneg hf h4

/-- Lemma B.6a (Derived Entropy-Rate Upper Bound `dHobs_dt_4D <= M_H_4D` on the Box): -/
theorem dHobs_dt_4D_le_M_H
    (c gamma_max H_max I_erase epsH : Real)
    (hc : 0 <= c) (hHmax : 0 <= H_max) (hIe : 0 <= I_erase)
    (s : ExistenceState)
    (hgamma_nonneg : 0 <= s.gamma) (hgamma_le : s.gamma <= gamma_max)
    (hH_nonneg : 0 <= s.H_obs)
    (htanh_H_nonneg : 0 <= Real.tanh (s.H_obs / epsH)) :
    dHobs_dt_4D c H_max I_erase epsH s <= M_H_4D c gamma_max H_max I_erase := by
  unfold dHobs_dt_4D M_H_4D
  have hc_gamma_nonneg : 0 <= c * s.gamma := mul_nonneg hc hgamma_nonneg
  have h1 : H_max - s.H_obs <= H_max := sub_le_self H_max hH_nonneg
  have h_step1 : c * s.gamma * (H_max - s.H_obs) <= c * s.gamma * H_max :=
    mul_le_mul_of_nonneg_left h1 hc_gamma_nonneg
  have h2 : c * s.gamma <= c * gamma_max := mul_le_mul_of_nonneg_left hgamma_le hc
  have h_step2 : c * s.gamma * H_max <= c * gamma_max * H_max :=
    mul_le_mul_of_nonneg_right h2 hHmax
  have h4 : 0 <= I_erase * Real.tanh (s.H_obs / epsH) := mul_nonneg hIe htanh_H_nonneg
  linarith

/-- Lemma B.6a' (Derived Entropy-Rate Lower Bound `-M_H_4D <= dHobs_dt_4D` on the Box):
On `B_4` (`0 <= s.gamma`, `s.H_obs <= H_max`, `Real.tanh (s.H_obs / epsH) <= 1`), the production term
`c * s.gamma * (H_max - s.H_obs)` is non-negative and the erasure sink is at most `I_erase <= M_H_4D`,
so `-M_H_4D <= -I_erase <= dHobs_dt_4D`. -/
theorem dHobs_dt_4D_ge_neg_M_H
    (c gamma_max H_max I_erase epsH : Real)
    (hc : 0 <= c) (hgmax : 0 <= gamma_max) (hHmax : 0 <= H_max) (hIe : 0 <= I_erase)
    (s : ExistenceState)
    (hgamma_nonneg : 0 <= s.gamma) (hH_le : s.H_obs <= H_max)
    (htanh_H_le_one : Real.tanh (s.H_obs / epsH) <= 1) :
    -M_H_4D c gamma_max H_max I_erase <= dHobs_dt_4D c H_max I_erase epsH s := by
  unfold dHobs_dt_4D M_H_4D
  have hc_gamma : 0 <= c * s.gamma := mul_nonneg hc hgamma_nonneg
  have hH_sub : 0 <= H_max - s.H_obs := sub_nonneg.mpr hH_le
  have h_prod : 0 <= c * s.gamma * (H_max - s.H_obs) := mul_nonneg hc_gamma hH_sub
  have h_erase : I_erase * Real.tanh (s.H_obs / epsH) <= I_erase * 1 :=
    mul_le_mul_of_nonneg_left htanh_H_le_one hIe
  have h_cmax : 0 <= c * gamma_max * H_max := mul_nonneg (mul_nonneg hc hgmax) hHmax
  linarith

/-- Theorem B.5c (Uniform Linear Comparison Lower Bound on `dIdot_dt_4D` for `liminf I_dot > 0`):
Using `0 <= tanh(I_dot / eps_I) <= I_dot / eps_I` and `-M_H_4D <= dHobs_dt_4D` on `B_4`, the processing rate obeys
`f P_0 - e ((I_dot / eps_I) M_H_4D) - g I_dot <= dIdot_dt_4D`, supplying the uniform linear differential lower
bound that forces `liminf_{t -> infty} I_dot(t) >= f P_0 / (g + e M_H_4D / eps_I) > 0` whenever `P_0 > 0`. -/
theorem dIdot_dt_4D_ge_linear_floor
    (e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max I_erase epsH epsI : Real)
    (hc : 0 <= c) (he : 0 <= e) (hf : 0 <= f)
    (hgmax : 0 <= gamma_max) (hHmax : 0 <= H_max) (hIe : 0 <= I_erase)
    (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (hMH : 0 <= M_H_4D c gamma_max H_max I_erase)
    (s : ExistenceState)
    (hgamma_nonneg : 0 <= s.gamma) (hH_le : s.H_obs <= H_max) (hPsi_nonneg : 0 <= s.Psi)
    (htanh_H_le_one : Real.tanh (s.H_obs / epsH) <= 1)
    (htanh_I_nonneg : 0 <= Real.tanh (s.I_dot / epsI))
    (htanh_I_le_lin : Real.tanh (s.I_dot / epsI) <= s.I_dot / epsI) :
    f * P_0 - e * ((s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase) - g * s.I_dot <=
      dIdot_dt_4D e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI s := by
  have hdH_lo : -M_H_4D c gamma_max H_max I_erase <= dHobs_dt_4D c H_max I_erase epsH s :=
    dHobs_dt_4D_ge_neg_M_H c gamma_max H_max I_erase epsH hc hgmax hHmax hIe s
      hgamma_nonneg hH_le htanh_H_le_one
  unfold dIdot_dt_4D P_eff
  have hdH_sum : 0 <= dHobs_dt_4D c H_max I_erase epsH s + M_H_4D c gamma_max H_max I_erase := by
    linarith
  have h_step1 :
      0 <= Real.tanh (s.I_dot / epsI) *
        (dHobs_dt_4D c H_max I_erase epsH s + M_H_4D c gamma_max H_max I_erase) :=
    mul_nonneg htanh_I_nonneg hdH_sum
  have h_step2 :
      Real.tanh (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase <=
      (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_right htanh_I_le_lin hMH
```

```
  have h_step3 :
      0 <= Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s +
        (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase := by
    linarith
  have h_step4 :
      0 <= e * (Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s +
        (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase) :=
    mul_nonneg he h_step3
  have h_assoc :
      e * Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s =
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) :=
    mul_assoc e (Real.tanh (s.I_dot / epsI)) (dHobs_dt_4D c H_max I_erase epsH s)
  have h_sr : 0 <= lam_sim * r_lrsc := mul_nonneg hsim hr
  have h_gam_term : 0 <= lam_sim * r_lrsc * s.gamma := mul_nonneg h_sr hgamma_nonneg
  have h_psi_term : 0 <= lam_lucid * s.Psi := mul_nonneg hluc hPsi_nonneg
  have h_Peff_lo : P_0 <= P_0 + lam_sim * r_lrsc * s.gamma + lam_lucid * s.Psi := by
    linarith
  have h_fPeff_lo : f * P_0 <= f * (P_0 + lam_sim * r_lrsc * s.gamma + lam_lucid * s.Psi) :=
    mul_le_mul_of_nonneg_left h_Peff_lo hf
  linarith

/-- Lemma B.6b (Upper Bound on Effective Flux `P_eff <= P_eff_max` on `B_4`): -/
theorem P_eff_le_P_eff_max
    (P_0 lam_sim r_lrsc gamma_max lam_lucid : Real)
    (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (s : ExistenceState) (hgamma_le : s.gamma <= gamma_max) (hPsi_le : s.Psi <= 1) :
    P_eff P_0 lam_sim r_lrsc lam_lucid s <= P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid := by
  unfold P_eff P_eff_max
  have h_sr_nonneg : 0 <= lam_sim * r_lrsc := mul_nonneg hsim hr
  have h_term1 : lam_sim * r_lrsc * s.gamma <= lam_sim * r_lrsc * gamma_max :=
    mul_le_mul_of_nonneg_left hgamma_le h_sr_nonneg
  have h_term2 : lam_lucid * s.Psi <= lam_lucid * 1 :=
    mul_le_mul_of_nonneg_left hPsi_le hluc
  linarith

/-- Theorem B.6 (Face 6 --- Upper I_dot-Ceiling Condition with Derived `dHobs_dt_4D <= M_H_4D`, I_dot =
     I_dot_max_4D):
Here `dHobs_dt_4D <= M_H_4D` is derived internally from box membership (`0 <= s.gamma <= gamma_max`,
`0 <= s.H_obs`) via `dHobs_dt_4D_le_M_H`. -/
theorem dIdot_dt_4D_nonpos_at_ceiling
    (e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max I_erase epsH epsI : Real)
    (hc : 0 <= c) (he : 0 <= e) (hf : 0 <= f) (hg : 0 < g)
    (hHmax : 0 <= H_max) (hIe : 0 <= I_erase)
    (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (hMH : 0 <= M_H_4D c gamma_max H_max I_erase)
    (s : ExistenceState)
    (hgamma_nonneg : 0 <= s.gamma) (hgamma_le : s.gamma <= gamma_max)
    (hH_nonneg : 0 <= s.H_obs) (hPsi_le : s.Psi <= 1)
    (hI : s.I_dot = I_dot_max_4D e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max I_erase)
    (htanh_H_nonneg : 0 <= Real.tanh (s.H_obs / epsH))
    (htanh_I_nonneg : 0 <= Real.tanh (s.I_dot / epsI))
    (htanh_I_le_one : Real.tanh (s.I_dot / epsI) <= 1) :
    dIdot_dt_4D e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI s <= 0 := by
  have hdH : dHobs_dt_4D c H_max I_erase epsH s <= M_H_4D c gamma_max H_max I_erase :=
    dHobs_dt_4D_le_M_H c gamma_max H_max I_erase epsH hc hHmax hIe s
      hgamma_nonneg hgamma_le hH_nonneg htanh_H_nonneg
  unfold dIdot_dt_4D I_dot_max_4D at *
  have hg_ne : g != 0 := ne_of_gt hg
  have h_g_mul :
      g * s.I_dot =
      e * M_H_4D c gamma_max H_max I_erase +
        f * P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid := by
    rw [hI]
    exact mul_div_cancel0
      (e * M_H_4D c gamma_max H_max I_erase +
        f * P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid) hg_ne
  have h_step1 :
      Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s <=
      Real.tanh (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left hdH htanh_I_nonneg
  have h_step2 :
      Real.tanh (s.I_dot / epsI) * M_H_4D c gamma_max H_max I_erase <=
      1 * M_H_4D c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_right htanh_I_le_one hMH
  have h_step3 :
      Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s <=
```

```
      M_H_4D c gamma_max H_max I_erase := by
    linarith
  have h_step4 :
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) <=
      e * M_H_4D c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left h_step3 he
  have h_assoc :
      e * Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s =
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt_4D c H_max I_erase epsH s) :=
    mul_assoc e (Real.tanh (s.I_dot / epsI)) (dHobs_dt_4D c H_max I_erase epsH s)
  have h_P_bound :
      P_eff P_0 lam_sim r_lrsc lam_lucid s <=
      P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid :=
    P_eff_le_P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid hsim hr hluc s hgamma_le hPsi_le
  have h_fP_bound :
      f * P_eff P_0 lam_sim r_lrsc lam_lucid s <=
      f * P_eff_max P_0 lam_sim r_lrsc gamma_max lam_lucid :=
    mul_le_mul_of_nonneg_left h_P_bound hf
  linarith

/-- Theorem B.7 (Face 7 --- Lower Psi-Boundary Condition with State-Dependent `R_self_sq`, Psi = 0): -/
theorem dPsi_dt_4D_nonneg_at_boundary
    (kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI : Real)
    (hkappa : 0 <= kappa_meta)
    (s : ExistenceState) (hPsi : s.Psi = 0) :
    0 <= dPsi_dt_4D kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI s := by
  unfold dPsi_dt_4D
  rw [hPsi, sub_zero, mul_one, mul_zero, sub_zero]
  have hR : 0 <= R_self_sq e f P_0 sigma_0 c H_max I_erase epsH epsI s :=
    R_self_sq_nonneg e f P_0 sigma_0 c H_max I_erase epsH epsI s
  exact mul_nonneg hkappa hR

/-- Theorem B.8 (Face 8 --- Upper Psi-Ceiling Condition with State-Dependent `R_self_sq`, Psi = 1): -/
theorem dPsi_dt_4D_neg_at_ceiling
    (kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI : Real)
    (hmu : 0 < mu_Psi)
    (s : ExistenceState) (hPsi : s.Psi = 1) :
    dPsi_dt_4D kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase epsH epsI s < 0 := by
  unfold dPsi_dt_4D
  rw [hPsi, sub_self, mul_zero, zero_sub, mul_one]
  exact neg_lt_zero.mpr hmu

/-- Theorem B.9 (Unified 8-Face 4D Box Pointwise Boundary Certificate):
Simultaneous formal verification of all eight pointwise boundary and ceiling sign conditions of
`B_4 = [0, gamma_max] x [0, H_max] x [0, I_dot_max_4D] x [0, 1]` with the explicit state-dependent
smooth quadratic `R_self_sq` and derived `dHobs_dt_4D <= M_H_4D`. -/
theorem existence_box_8face_nagumo_certificate
    (a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
     I_erase epsH epsI kappa_meta mu_Psi : Real)
    (ha : 0 <= a) (hb : 0 < b) (hc : 0 <= c) (he : 0 <= e) (hf : 0 < f) (hg : 0 < g)
    (hgmax : 0 < gamma_max) (hHmax : 0 < H_max) (hE : 0 <= E_avail) (hP0 : 0 < P_0)
    (hsim : 0 <= lam_sim) (hr : 0 <= r_lrsc) (hluc : 0 <= lam_lucid)
    (hIe : 0 < I_erase) (htanh_Hmax : 0 < Real.tanh (H_max / epsH))
    (hMH : 0 <= M_H_4D c gamma_max H_max I_erase)
    (hkappa : 0 <= kappa_meta) (hmu : 0 < mu_Psi) :
    -- Face 1: Lower gamma-boundary (gamma = 0)
    (forall s : ExistenceState, s.gamma = 0 -> 0 <= s.I_dot ->
      0 <= (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
            I_erase epsH epsI kappa_meta mu_Psi s).gamma) /\
    -- Face 2: Upper gamma-ceiling (gamma = gamma_max)
    (forall s : ExistenceState, s.gamma = gamma_max ->
      (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
        I_erase epsH epsI kappa_meta mu_Psi s).gamma < 0) /\
    -- Face 3: Lower H_obs-boundary (H_obs = 0)
    (forall s : ExistenceState, s.H_obs = 0 -> 0 <= s.gamma ->
      0 <= (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
            I_erase epsH epsI kappa_meta mu_Psi s).H_obs) /\
    -- Face 4: Upper H_obs-ceiling (H_obs = H_max)
    (forall s : ExistenceState, s.H_obs = H_max ->
      (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
        I_erase epsH epsI kappa_meta mu_Psi s).H_obs < 0) /\
    -- Face 5: Lower I_dot-boundary (I_dot = 0)
    (forall s : ExistenceState, s.I_dot = 0 -> 0 <= s.gamma -> 0 <= s.Psi ->
      0 < (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
            I_erase epsH epsI kappa_meta mu_Psi s).I_dot) /\
    -- Face 6: Upper I_dot-ceiling (I_dot = I_dot_max_4D, with dHobs_dt_4D <= M_H_4D derived from box membership)
```

```
    (forall s : ExistenceState,
      0 <= s.gamma -> s.gamma <= gamma_max -> 0 <= s.H_obs -> s.Psi <= 1 ->
      s.I_dot = I_dot_max_4D e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max I_erase ->
      0 <= Real.tanh (s.H_obs / epsH) ->
      0 <= Real.tanh (s.I_dot / epsI) ->
      Real.tanh (s.I_dot / epsI) <= 1 ->
      (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
        I_erase epsH epsI kappa_meta mu_Psi s).I_dot <= 0) /\
    -- Face 7: Lower Psi-boundary (Psi = 0, state-dependent R_self_sq >= 0 derived automatically)
    (forall s : ExistenceState, s.Psi = 0 ->
      0 <= (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
            I_erase epsH epsI kappa_meta mu_Psi s).Psi) /\
    -- Face 8: Upper Psi-ceiling (Psi = 1)
    (forall s : ExistenceState, s.Psi = 1 ->
      (F_exist a b c e f g gamma_max H_max E_avail P_0 sigma_0 lam_sim r_lrsc lam_lucid
        I_erase epsH epsI kappa_meta mu_Psi s).Psi < 0) := by
  refine <?_, ?_, ?_, ?_, ?_, ?_, ?_, ?_>
  . intro s hgamma hI
    exact dgamma_dt_4D_nonneg_at_boundary a b E_avail gamma_max ha hE s hgamma hI
  . intro s hgamma
    exact dgamma_dt_4D_neg_at_ceiling a b E_avail gamma_max hb hgmax s hgamma
  . intro s hH hgamma
    exact dHobs_dt_4D_nonneg_at_boundary c H_max I_erase epsH hc (le_of_lt hHmax) s hH hgamma
  . intro s hH
    exact dHobs_dt_4D_neg_at_ceiling c H_max I_erase epsH hIe htanh_Hmax s hH
  . intro s hI hgamma hPsi
    exact dIdot_dt_4D_pos_at_boundary e f g P_0 lam_sim r_lrsc lam_lucid c H_max I_erase epsH epsI
      hf hP0 hsim hr hluc s hI hgamma hPsi
  . intro s hgamma_nonneg hgamma_le hH_nonneg hPsi_le hI htH htI0 htI1
    exact dIdot_dt_4D_nonpos_at_ceiling e f g P_0 lam_sim r_lrsc lam_lucid c gamma_max H_max
      I_erase epsH epsI hc he (le_of_lt hf) hg (le_of_lt hHmax) (le_of_lt hIe) hsim hr hluc hMH
      s hgamma_nonneg hgamma_le hH_nonneg hPsi_le hI htH htI0 htI1
  . intro s hPsi
    exact dPsi_dt_4D_nonneg_at_boundary kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase
      epsH epsI hkappa s hPsi
  . intro s hPsi
    exact dPsi_dt_4D_neg_at_ceiling kappa_meta mu_Psi e f P_0 sigma_0 c H_max I_erase
      epsH epsI hmu s hPsi

#print axioms R_self_sq_nonneg
#print axioms dgamma_dt_4D_ge_neg_decay
#print axioms dHobs_dt_4D_ge_neg_M_H
#print axioms dIdot_dt_4D_ge_linear_floor
#print axioms dIdot_dt_4D_nonneg_at_boundary_zero_drive
#print axioms existence_box_8face_nagumo_certificate

end
```

8 Six-Layer Verification Architecture & Quantitative Reproducibility Digest

Table 6 summarizes the six verification layers executed by `verification_existence_v1_0.py` and the exact mathematical or numerical scope of each layer.

Quantitative Archival Reproducibility Digest: The SHA-256 digest below hashes the computed exact rational Hopf thresholds $(377/84, 12069/4025, 97461/364100)$, the positivity domain $\mathcal{P}_{\text{crit}} = 10/99$, the primitive polynomial coefficients of $de_c/d\mathcal{P}_0$, the bisected $\mathcal{P}_0 = 0$ SNLC thresholds $(\lambda_{\text{lucid,SNLC}}(0) = 0.080341, \lambda_{\text{lucid,SNLC}}(0.80) = 0.057974)$, and the SHA-256 digests of `ExistenceVerification.lean` (17 theorems) and `lakefile.lean`:

`de998b2fa755830ef45864614776390eba573aa1323c61a9d8c65f1c0688bd18`

Table 6: Six-layer verification architecture for Appendix B v1.0 (`verification_existence_v1_0.py`).

Verification Layer	**Method / Tool**	**Verified Mathematical / Numerical Result & Scope Boundary**
1. Rational Hopf Locus & Domain ($\mathcal{P}_0 > 10/99$)	Exact `Fraction` & `SymPy`	Exact $e_c(5) = 377/84$, $e_c(4) = 12069/4025$, $e_c(1) = 97461/364100$; proves $H^*_{\text{obs,v1.4}} > 0 \iff \mathcal{P}_0 > 10/99 \approx 0.1010$ ($\epsilon \to 0^+$ limit, accurate at $\epsilon = 0.05$ for $\mathcal{P}_0 \geq 0.105$) and $de_c/d\mathcal{P}_0 > 0$ via positivity of all 8 primitive numerator and 7 denominator polynomial coefficients.
2. 4D `SymPy` Jacobian, $\mathcal{P}_0 = 0$ Origin & Variational $J_4(x)$	Symbolic `SymPy`	Proves $R_{\text{self}}\|_* = 0$ and $\nabla R_{\text{self}}\|_* = (0,0,0,0)$ at any steady state ($\dot{H}^*_{\text{obs}} = 0$), row 4 of J_4 is $[0,0,0,-\mu_\Psi]$, closed-loop $e_c^{\text{CL}}(\mathcal{P}_0, \Lambda_{\text{eff}})$ for $\mathcal{P}_0 > 0$, non-existence of positive interior equilibria at $\mathcal{P}_0 = 0, \Lambda_{\text{eff}} < 1$ (roots $\gamma^* \in \{0, -2.5\}$ at $\Lambda_{\text{eff}} = 0.80$), local stability of $(0,0,0,0)$, and $\max\|J_{\text{analytic}}(x) - J_{\text{SymPy}}(x)\| = 1.65 \times 10^{-17}$.
3. 8-Face Box Check, Uniform Floors & Poincaré Integrals	`SciPy LSODA` & `DOP853`	Verifies all 8 face inequalities on $\mathcal{B}_4$, uniform positive floors $\dot{I}_{\text{floor}}(\mathcal{P}_0) > 0$ and $\gamma_{\text{floor}}(\mathcal{P}_0) > 0$ for $\mathcal{P}_0 > 0$, `LSODA` vs. `DOP853` agreement on $\Sigma = \{\gamma = 2.0, \dot{\gamma} > 0\}$ across rtol $\in \{10^{-8}, 10^{-10}, 10^{-12}\}$, and whole-period integrals across Regimes I–IVc.
4. 20D Variational Floquet, 2×2 Ablation & SNLC	20D Monodromy ODE & Bisection	Integrates $\dot{\Phi} = J_4(x)\Phi(t)$ over one period T, verifying $\|\Phi(T)F - F\|/\|F\| \leq 1.72 \times 10^{-9}$, saltation-projected $\rho_{\text{var}}(DP_{2,H\dot{I}}) < 10^{-9}$, $\mu_{\Psi,\text{exact}} = 2.537698 \times 10^{-6}$ in Regime II, 2×2 zero-drive ablation (showing Ψ-memory sustains the $\mathcal{P}_0 = 0$ cycle while memoryless $\Lambda_{\text{eff}}\gamma$ collapses at the $\gamma_{\min} \approx 3.6 \times 10^{-4}$ trough), and SNLC thresholds $\lambda_{\text{lucid,SNLC}}(0) = 0.080341 \to \lambda_{\text{lucid,SNLC}}(0.80) = 0.057974$.
5. Minimum & Maximum Parameter Cube Corners	`brentq`, `LSODA` & `DOP853`	Verifies Minimum Corner ($e = 0.10, \mathcal{P}_0 = 0.10 < 10/99$, soft-wall root $H^*_{\text{sw}} = 0.1295$, $e_c^{\text{CL}} = 0.099132$, solver diff $< 10^{-8}$ at rtol $= 10^{-12}$) and Maximum Corner ($\gamma^* = (-11 + \sqrt{1471})/5$, $\mathcal{P}^*_{\text{eff}} = (32 + 2\sqrt{1471})/9$, $e_c^{\text{CL}} = 21.958110$).
6. Lean 4 Pointwise Algebra (17 Thms) & Reproducibility Hash	Lean 4 Lake (`lake build`) & Static Check	Offline source check of all 17 pointwise theorems in `ExistenceVerification.lean` (`0 sorry`, `0 nlinarith`, full compilation via `lake build` in GitHub Actions CI) and quantitative reproducibility digest below.

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